

Taste triggers a homeostatic temperature control in hungry flies

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Abstract

Summary

Hungry animals consistently show a desire to obtain food. Even a brief sensory detection of food can trigger bursts of physiological and behavioral changes. However, the underlying mechanisms by which the sensation of food triggers the acute behavioral response remain elusive. We have previously shown in *Drosophila* that hunger drives a preference for low temperature. Because *Drosophila* is a small ectotherm, a preference for low temperature implies a low body temperature and a low metabolic rate. Here, we show that taste sensing triggers a switch from a low to a high temperature preference in hungry flies. We show that taste stimulation by artificial sweeteners or optogenetics triggers an acute warm preference, but is not sufficient to reach the fed state. Instead, nutrient intake is required to reach the fed state. The data suggest that starvation recovery is controlled by two components: taste-evoked and nutrient-induced warm preferences, and that taste and nutrient quality play distinct roles in starvation recovery. Animals are motivated to eat based on time of day or hunger. We found that clock genes and hunger signals profoundly control the taste-evoked warm preferences. Thus, our data suggest that the taste-evoked response is one of the critical layers of regulatory mechanisms representing internal energy homeostasis and metabolism.

eLife assessment

This paper presents **valuable** findings that gustation and nutrition might independently influence the preferred environmental temperature in flies. The evidence supporting the main claims is **solid** and well presented. The finding that flies might thus exhibit a cephalic phase response similar to mammals will be of value for future investigations.

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Introduction

Animals are constantly sensing environmental stimuli and changing their behavior or physiology based on their internal state ^{1–7}. Hungry animals are strongly attracted to food. Immediately after seeing, smelling, or chewing food, even without absorbing nutrients, a burst of physiological changes is suddenly initiated in the body. These responses are known in mammals as the cephalic phase response (CPR) ⁸. For example, a flood of saliva and gastrointestinal secretions prepares hungry animals to digest food ^{9–13}. Starvation results in lower body temperatures, and chewing food triggers a rapid increase in heat production, demonstrating CPR in thermogenesis ^{14–16}. However, the underlying mechanisms of how the sensation of food without nutrients triggers the behavioral response remain unclear.

To address this question, we used a relatively simple and versatile model organism, *Drosophila melanogaster*. Flies exhibit robust temperature preference behavior ^{17,18}. Due to the low mass of small ectotherms, the source of temperature comes from the environment. Therefore, their body temperatures are close to the ambient temperature ^{19,20}. For temperature regulation, animals are not simply passive receivers of ambient temperature. Instead, they actively choose an ambient temperature based on their internal state. For example, we have shown that preferred temperature increases during the daytime and decreases during the night time, exhibiting circadian rhythms of temperature preference (temperature preference rhythms: TPR) ²¹. Because their surrounding temperature is very close to their body temperature, TPR leads to body temperature rhythms (BTR) that is very similar to mammalian BTR ^{22,23}. Another example is the starvation. Hunger stress forces flies to change their behavior and physiological response ^{24,25}. We previously showed that the hungry flies prefer a lower temperature ²⁶. The flies in a lower environmental temperature has been shown to have a lower metabolic rate, and the flies in a higher environmental temperature has been shown to have a higher metabolic rate ^{26–28}. Therefore, hungry flies choose a lower temperature and therefore, their metabolic rate is lower. Similarly, in mammals, starvation causes a lower body temperature, hypothermia ⁶. In mammals, body temperature is controlled by the balance between heat loss and heat production. The starved mammals have been shown a lower heat production ^{5–7}. Therefore, both flies and mammals, the starvation causes a low body temperature.

The flies exhibit robust feeding behaviors ^{29,30} and molecular and neural mechanisms of taste are well documented ^{2,31–36}. Therefore, we focused on taste and temperature regulation and asked how the taste cue triggers a robust behavioral recovery of temperature preference in starving flies. We show in hungry flies that taste without nutrients induces a switch from a low to a high temperature preference. While taste leads to a warmer temperature preference, nutrient intake causes the flies to prefer an even warmer temperature. This nutrient-induced warm preference results in a complete recovery from starvation. Thus, taste-evoked warm preference is different from nutrient-induced warm preference and potentially similar physiology as CPR. Therefore, when animals emerge from starvation, they use a two-step approach to recovery: taste-evoked and nutrient-induced warm preference. While a rapid component is elicited by food taste alone, a slower component requires nutrient intake.

Animals are motivated to eat based on their internal state, such as time of day or degree of hunger. The circadian clock drives daily feeding rhythms ^{37–39} and anticipates meal timing. Daily feeding timing influences energy homeostasis and metabolism ^{40,41}. Circadian clocks control feeding behavior in part via orexigenic peptidergic/hormonal regulation such as neuropeptide Y (NPY) and agouti-related peptide (AgRP) neurons, which are critical for regulating feeding and metabolism ^{39,42}. Feeding-fasting cycles modulate peripheral organs in liver, gut, pancreas, and so on ⁴³, suggesting that circadian clocks, peptidergic/hormonal signals, and peripheral organs are organically coordinated to enable animals maintain their internal states constantly. We

found that clock genes and hunger signals are strongly required for taste-evoked warm preference. The data suggest that taste-evoked response is an indispensable physiological response that represents internal state. Taken together, our data shed new light on the role of the taste-evoked response and highlight a crucial aspect of our understanding of feeding state and energy homeostasis.

Results

Food detection triggers a warm preference

To investigate how food detection influences temperature preference behavior in *Drosophila* ^{17,18}, the *white*¹¹¹⁸ (*w*¹¹¹⁸) control flies were fed fly food containing carbohydrate, protein, and fat sources (Fig. 1A ^{26,44} and S1 ^{26,44}) and tested in temperature preference behavioral assays (Fig. S1 ^{26,44}). The flies are released into a chamber set to a temperature range of 16–34°C and subsequently accumulate at their preferred temperature (Tp) at 25.2±0.2 °C within 30 minutes (min) (Fig. 1B ^{26,44}: fed, white bar). On the other hand, when *w*¹¹¹⁸ flies were starved overnight with water only, they preferred 21.7±0.3 °C (Fig. 1A ^{26,44} and 1B ^{26,44}: overnight starvation (STV), gray bar). Thus, starvation leads to a lower Tp. As we have previously reported, starvation strongly influences temperature preference ^{26,44}.

To examine how starved flies recover from lower Tp, they were offered fly food for 5 min, 10 min, 30 min, and 1 hr (Fig. 1A ^{26,44}). Immediately after the flies were refed, the temperature preference behavior assay was examined. The assay takes 30 min from the time the flies are placed in the apparatus until the final choice is made. For example, in the case of 5 min refed flies, it took a total of 35 min from the start of refeeding to the end of the assay. After 10 min, 30 min, or 1 hr of fly food refeeding, starved flies preferred a temperature similar to that of fed flies (Fig. 1B ^{26,44}, 1F ^{26,44}, and S2 ^{26,44}: statistics shown as red stars, Table S1), suggesting a full recovery from starvation. On the other hand, refeeding after 5 min resulted in a warmer temperature than the starved flies. Nevertheless, Tp did not reach that of the fed flies (Fig. 1B ^{26,44}, 1F ^{26,44}, and S2 ^{26,44}). Therefore, refeeding the flies for 5 min resulted in a partial recovery from the starved state (Fig. 1B ^{26,44}, 1F ^{26,44}, and S2 ^{26,44}, statistics shown as green stars, Table S1). Thus, our data suggest that food intake triggers a warm preference in starved flies.

Sucralose refeeding promotes a warm preference

While only 5 min of refeeding fly food caused hungry flies to prefer a slightly warmer temperature, 10 min of refeeding caused hungry flies prefer a similarly warmer temperature as the fed flies (Fig. 1B ^{26,44}, 1F ^{26,44}, and S2 ^{26,44}). Therefore, we hypothesized that food-sensing cues might be important for the warm preference. Sucralose is an artificial sweetener that activates sweet taste neurons in *Drosophila* ⁴⁵ and modulates taste behaviors such as the proboscis extension reflex ^{46–49}. Importantly, sucralose is a non-metabolizable sugar and has no calories. Therefore, to investigate how food-sensing cues are involved in warm preference, we examined how sucralose refeeding changes the temperature preference of starved flies. After starved flies were refed sucralose for 10 min or 1 hr (Fig. 1A ^{26,44}), they preferred a warmer temperature; however, Tp was halfway between Tps of fed and starved flies (Fig. 1C, F ^{26,44} and S2 ^{26,44}: blue, Table S1). Thus, sucralose ingestion induces a warm preference but shows a partial recovery of Tp from the starved state.

While refeeding the flies with food resulted in a full recovery of Tp from the starved state, refeeding them with sucralose resulted in only a partial recovery of Tp. Therefore, starved flies may use both taste cues and nutrients to fully recover Tp from the starved state. To evaluate this possibility, we used glucose, which contains both sweetness (gustatory cues) and nutrients (i.e., metabolizable sugars) (Fig. 1A ^{26,44}), and tested glucose refeeding for 5 min, 10 min, and 1 hr. We found that 10 min or 1 hr glucose refeeding resulted in full recovery of Tp from the starved state

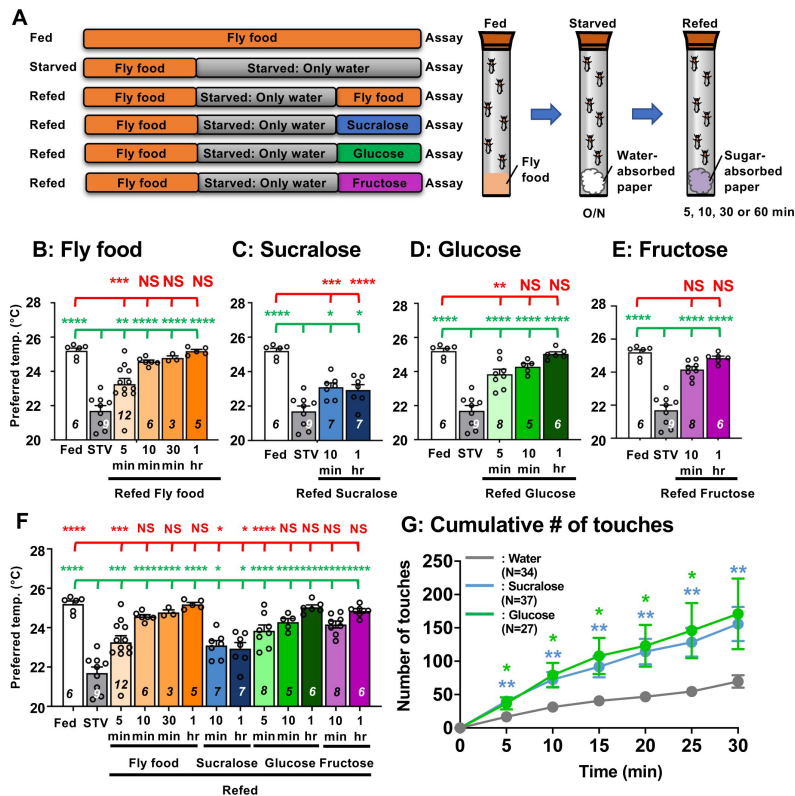


Figure 1.

Hungry flies switch from cold to warm preference upon food detection.

(A) A schematic diagram of the experimental conditions. Flies were reared on fly food. Emerging flies were kept on fly food for one or two days (orange) and then starved with water-absorbing paper. Starvation was applied for 1-3 overnights (ON). The duration of starvation was chosen when the flies showed a statistically significant difference in Tp between fed and starved conditions (see "Starvation conditions" in Materials and methods for details). Starved flies were refeed with either fly food (orange bar), 2 mL sucralose (blue), glucose solution (green), or fructose solution (purple) absorbed paper and then examined for temperature preference (Tp) behavioral assays. Details are provided in the experimental procedures.

(B-E) Comparison of preferred temperature (Tp) of *white¹¹¹⁸* (*w¹¹¹⁸*) control flies between fed (white bar), starved (STV; gray bar), and refeed (orange, blue, or green bar) states. Starvation was applied for 1ON. Starved flies were refeed with fly food (orange bar) for 5, 10, 30, or 60 min (1 hr) (B), 2.8 mM sucralose solution (blue bar) for 10 min or 1 hr (C), 2.8 mM (equivalent to 5%) glucose solution (green bar) for 5 min, 10 min, or 1 hr (D), or 2.8mM fructose solution (purple bar) for 10 min or 1 hr (E). Behavioral experiments were performed at the specific time points, Zeitgeber time (ZT) 4-7. ZT0 and ZT12 are light on and light off, respectively. Dots on each bar indicate individual Tp in the assays. Numbers in italics indicate the number of trials. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA was performed to compare Tp between each refeeding condition. Red or green stars indicate Tukey's post hoc test comparing differences between experimental and fed or starved conditions, respectively. Data are presented as mean Tp with SEM. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001. NS indicates no significance.

(F) Comparison of Tp among the following conditions in *w¹¹¹⁸* (control) flies: fed (white bar), starved overnight (O/N) (water or STV; gray bar), refeed with fly food (fly food; orange bar), sucralose (blue bars), glucose (green bars), or fructose (purple bars). The same data are used in **Fig. 1B-E**. A distribution of temperature preference in each condition is shown in **Fig. S2A**. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA was performed to compare Tp between each refeeding condition. Red (or green) stars indicate Tukey's post hoc test comparing differences between experimental and fed (or starved) conditions, respectively. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001. NS indicates not significant.

(G) Feeding assay: The number of touches to water (gray), 2.8 mM (equivalent to 5%) glucose (green), or 2.8 mM sucralose in each solution (blue) was examined using *w¹¹¹⁸* flies starved for 24 hrs. Water, glucose and sucralose were tested individually in the separate experiments. A cumulative number of touches to water or sugar solution for 0-30 min was plotted. Two-way ANOVA was used for multiple comparisons. Blue and green stars show Fisher's LSD post hoc test comparing sucralose (blue stars) or glucose (green stars) solution feeding to water drinking. All data shown are means with SEM. **p* < 0.05. ***p* < 0.01. ****p* < 0.001. *****p* < 0.0001.

(Fig. 1D, 1F and S2: statistics shown in red NS, Table S1) and was significantly different from starved flies (Fig. 1D, 1F and S2: statistics shown in green stars, Table S1). Thus, our data showed that sucralose refeeding induced partial recovery and glucose refeeding induced full recovery from the starved state.

It is still possible that the starved flies consumed glucose faster than sucralose during the first 10 min, which could result in a different warming preference. To rule out this possibility, we examined how often starved flies touched glucose, sucralose, or water during the 30 min using the Fly Liquid-food Interaction Counter (FLIC) system⁵⁰. The FLIC system assays allow us to monitor how much interaction between the fly and the liquid food reflects feeding episodes. We found that starved flies touched glucose and sucralose food at similar frequencies and more frequently than water during the 30 min test period (Fig. 1G, Table S1). The data suggest that flies are likely to feed on glucose and sucralose at similar rates. Therefore, we concluded that the differential effect of sucralose and glucose refeeding on temperature preference was not due to differences in feeding rate.

Furthermore, to confirm that sweet taste is more important than sugar structure, we instead used fructose, another simple sugar that contains sweetness and nutrients. Glucose and fructose are monosaccharides and a member of hexose and pentose, respectively. We tested fructose refeeding for 10 min and 1 hr. We found that 10 min of fructose refeeding resulted in full recovery of Tp from the starved state (Fig. 1E, 1F and S2: statistics shown in red NS, Table S1), and 10 min and 1 hr fructose refeeding were significantly different from starved flies (Fig. 1E, 1F and S2: statistics shown in green stars, Table S1). The data suggest that sweet taste is more important than the structures of the sugar compounds.

Activation of sweet taste neurons leads to warm preference

To determine how taste elicits a warm preference, we focused on the sweet gustatory receptors (Grs), which detect sweet taste. We used sweet Gr mutants and asked whether sweet Grs are involved in taste-evoked warm preference. Two different sweet Gr mutants, *Gr5a*^{-/-}; *Gr64a*^{-/-} and *Gr5a*^{-/-}; *Gr61a*^{-/-}; *Gr64a*^{-/-}, are known to reduce sugar sensitivity compared to the control^{47,48,51,52}. We found that sweet Gr mutant flies exhibited a normal starvation response in which the Tp of starved flies was lower than that of fed flies (Fig. 2A and 2B, white and gray bars, statistics shown as green stars, Table S1). However, starved sweet Gr mutant flies did not increase Tp after 10 min sucralose refeeding (Fig. 2A and 2B, blue bars, statistics shown as green and red stars, Table S1). These data suggest that sweet Grs are involved in taste-evoked warm preference.

Sweet Grs are expressed in the sweet Gr-expressing neurons (GRNs) located in the proboscis and forelegs^{52,53}. To determine whether sweet GRNs are involved in taste-evoked warm preference, we silenced all sweet GRNs. We expressed the inwardly rectifying K⁺ channel *Kir2.1* (*uas-Kir*)⁵⁴ using *Gr64f-Gal4*, which is expressed in all sweet GRNs in the proboscis and forelegs^{47,52,53}. Inactivation of all sweet GRNs showed a normal starvation response (Fig. 2C, white and gray bars, statistics shown as green stars). However, flies silencing all sweet GRNs failed to show a warm preference after 10 min of sucralose refeeding (Fig. 2C, blue bar, statistics shown as green and red stars, Table S1). This phenotype was similar to the data obtained with the sweet Gr mutant strains (Fig. 2A and 2B). On the other hand, control flies (*Gr64f-Gal4*⁺ and *uas-Kir*⁺) showed a normal starvation response and a taste-evoked warm preference (Fig. 2D and 2E, gray and blue bars, respectively, statistics shown as green and red stars, Table S1). Thus, our data indicate that sweet GRNs are required for taste-evoked warm preference.

To further investigate whether activation of sweet GRNs induces a warm preference, we used the optogenetic approach, a red light sensitive channelrhodopsin, CsChrimson^{55,56}. Starved flies were given water containing 0.8 mM all-trans-retinal (ATR), the chromophore required for CsChrimson activation. These flies were not fed sucralose; instead, gustatory neurons in starved

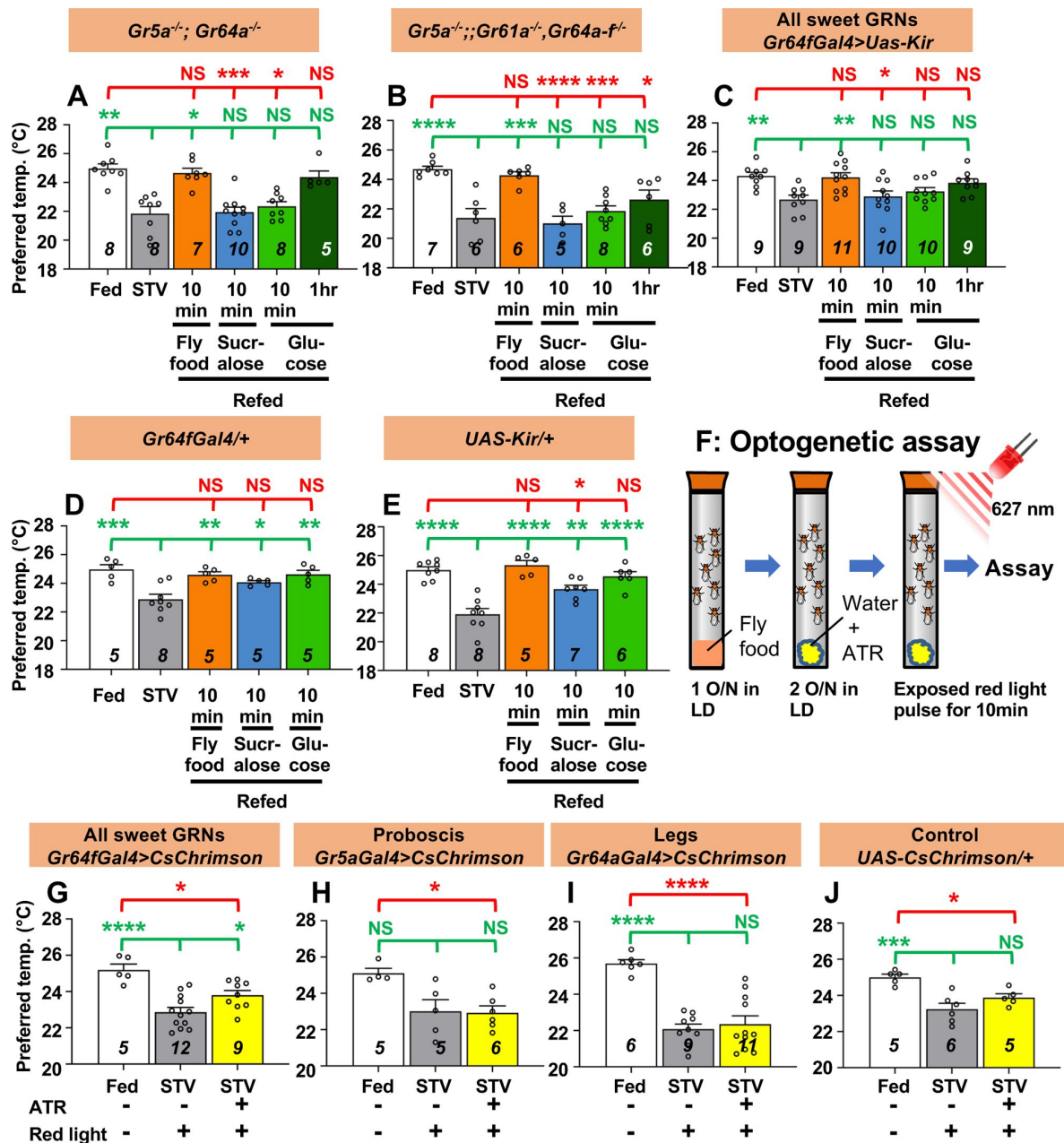


Figure 2.

Gustatory neurons are essential for taste-evoked warm preference.

(A-E) Comparison of preferred temperature (Tp) of flies between fed (white bar), starved (STV; gray bar), and refeed (orange, blue, green, or dark green bar) states. Flies were starved for 2 overnights (ON) except for *Gr5a^{-/-}; Gr61a^{-/-}; Gr64a^{-/-}* (1.5ON). Starved flies were refeed with fly food for 10 min (fly food; orange bars), sucrose for 10 min (blue bars), or glucose for 10 min (green bars), or 1 hr (dark green bars).

(F) Schematic of the optogenetic activation assay.

(G-J) Comparison of Tp of flies between fed (white bar), starved (STV; gray bar), and starved with all-trans-retinal (ATR; yellow bars), which is the chromophore required for CsChrimson activation. Gustatory neurons in starved flies were excited by red light pulses (flashing on and off at 10 Hz) for 10 min. Starvation was performed for 2ON. Averaged fly distributions of flies in the temperature gradient for *Gr64fGal4>CsChrimson* were shown in Fig. S2B. Behavioral experiments were performed on ZT4-7. Dots on each bar indicate individual Tp in assays. Numbers in italics indicate the number of trials. The Shapiro-Wilk test was used to test for normality. One-way ANOVA was used for statistical analysis. Red or green stars indicate Tukey's post hoc test compared between each experiment and the fed (red) or starved (green) condition. All data presented are means with SEM. *p < 0.05. **p < 0.01. ***p < 0.001. ****p < 0.0001. NS indicates not significant.

flies were excited by red light pulses (flashing on and off at 10 Hz) for 10 min (**Fig. 2F**). In this case, although the flies were not refed, the gustatory neurons were artificially excited by CsChrimson activation so that we could evaluate the effect of excitation of sweet GRNs on taste-evoked warm preference.

CsChrimson was expressed in sweet GRNs in the proboscis and legs (all sweet GRNs) using *Gr64f-Gal4*. These flies showed a normal starvation response (**Fig. 2G–2J**, white and gray bars, statistics shown as green and red stars, Table S1). Excitation of all sweet GRNs by red light pulses elicited a warm preference, and Tp was intermediate between fed and starved flies, suggesting partial recovery (**Fig. 2G**, yellow bar, statistics shown as green and red stars, **Fig. S2B**, Table S1). However, neither excitation of *Gr5a*- (**Fig. 2H**) nor *Gr64a*-expressing neurons (**Fig. 2I**) induced a warm preference (yellow bars, Table S1). While *Gr64a-Gal4* is expressed only in the legs, *Gr5a-Gal4* is expressed in the proboscis and legs, but does not cover all sweet GRNs like *Gr64f-Gal4*. Notably, control flies (*UAS-CsChrimson/+*) did not show a warm preference to red light pulses with ATR application (**Fig. 2J**). Taken together, our data suggest that excitation of all sweet GRNs results in a warm preference.

We next asked whether the sweet Grs contribute to the nutrient-induced warm preference. We found that all these starved flies did not increase Tp after 10 min of glucose intake (**Fig. 2A–2C**, green bars, statistics shown as green and red stars, Table S1). All control flies showed normal responses to 10 min of glucose refeeding (**Fig. 2D** and **2E**, green bars, Table S1). The data suggest that the sweet Grs which we tested are potentially expressed in tissues/neurons required for internal nutrient sensing. Notably, we found that flies increased Tp after 10 min of refeeding with fly food containing carbohydrate, fat, and protein (**Fig. 2A–2C**, orange bars, statistics shown as green and red stars, Table S1). All control flies showed normal responses to 10 min of fly food intake (**Fig. 2D** and **2E**, orange bars, Table S1). The data suggest that gustatory neurons are required for warm preference in carbohydrate refeeding, but not for other nutrients such as fat or protein (see Discussion). Because flies have sensory neurons that detect fatty acids or amino acids, these neurons may drive the response to fly food intake. This is likely why the sweet-insensitive flies can still recover after eating fly food (**Fig. 2A–2C**, orange bars).

The temperature-sensing neurons are involved in taste-evoked warm preference

The warm-sensing neurons, anterior cells (ACs), and the cold-sensing *R11F02-Gal4*-expressing neurons control temperature preference behavior. Small ectotherms such as *Drosophila* set their Tp to avoid noxious temperatures using temperature information from cold- and warm-sensing neurons. We have previously shown that starved flies choose a lower Tp, the so-called hunger-driven lower Tp. ACs control the hunger-driven lower Tp, but cold-sensing *R11F02-Gal4*-expressing neurons do not. ACs express transient receptor potential A1 (TrpA1), which responds to a warm temperature >25 °C. The set point of ACs in fed flies, which is ~25 °C, is lowered in starved flies. Therefore, the lower set point of ACs corresponds to the lower Tp in starved flies.

First, we asked whether ACs are involved in taste-evoked warm preference. Because the ACs are important for the hunger-driven lower Tp, the AC-silenced flies did not show a significant difference in Tp between fed and starved conditions for only one overnight of starvation. Therefore, we first extended the starvation time to two overnights so that the AC-silenced flies showed a significant difference in Tp between fed and starved conditions (**Fig. 3A**, white and gray bars, statistics shown as green and red stars, Table S1). Importantly, longer periods of starvation do not affect the ability of *w¹¹¹⁸* flies to recover (**Fig. S3**).

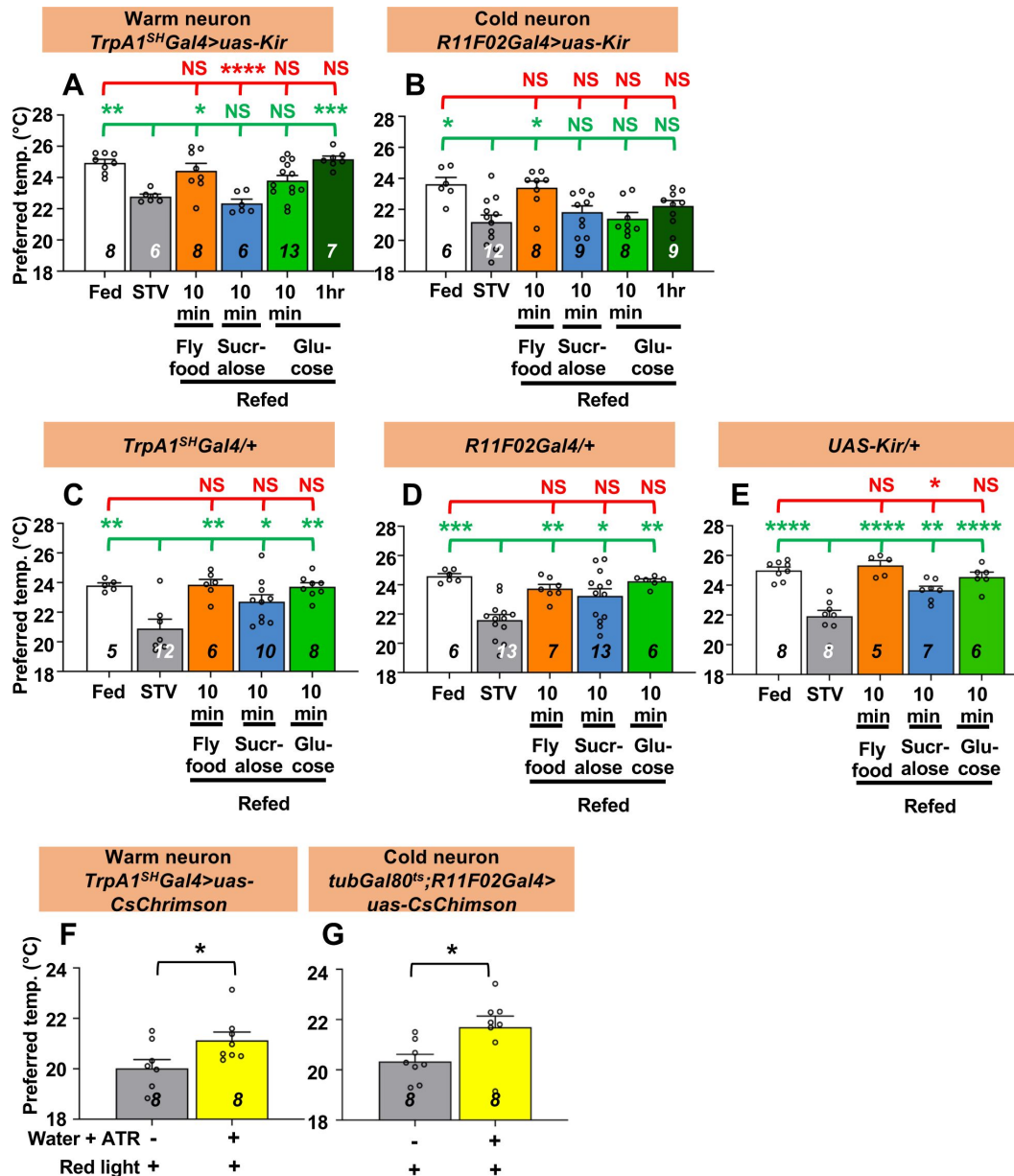


Figure 3.

Both warm and cold temperature sensing neurons are involved in taste-evoked warm preference.

(A-E) Comparison of preferred temperature (Tp) of flies between fed (white bar), starved (STV; gray bar), and refeed (orange, blue, green, or dark green bar) conditions. Starvation was applied for 2 overnights (ON). Starved flies were refeed with fly food for 10 min (orange bar), sucralose for 10 min (blue bar), or glucose for 10 min (green bar) or 1 hr (dark green bar). The Shapiro-Wilk test was used to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Red or green stars indicate Tukey's post hoc test or Dunn's test compared between each experiment and the fed (red) or starved (green) condition, respectively.

(F and G) Comparison of Tp between starved (STV; gray bar) and all-trans-retinal (ATR; yellow bar) starved flies. Starvation was performed for 2ON. Warm neurons (F) or cold neurons (G) in starved flies expressed CsChrimson, which was excited by red light pulses for 10 min. The Shapiro-Wilk test was performed to test for normality. Student's t-test or Kolmogorov-Smirnov test was used for statistical analysis. (G) *tubGal80^{TS}; R11F02-Gal4>uas-CsChrimson* flies were reared at 18°C, and emerged adults were collected and stored at 29°C. See Materials and Methods for details. These behavioral experiments were performed on ZT4-7. The dots on each bar indicate individual Tp in the assays. Numbers in italics indicate the number of experiments. All data shown are means with SEM. *p < 0.05. **p < 0.01. ***p < 0.001. ****p < 0.0001. NS

To examine whether ACs regulate taste-evoked warm preference, we refed sucralose to AC-silenced flies for 10 min. We found that the Tp of the refed flies was still similar to that of the starved flies (**Fig. 3A**, blue bar, statistics shown as green and red stars, Table S1), indicating that sucralose refeeding could not restore Tp and that ACs are involved in taste-evoked warm preference. Significantly, even when the AC-silenced flies were starved for two overnights, they were able to recover Tp to the normal food for 10 min and to glucose for 1 hr (**Fig. 3A**, orange and green bars), suggesting that the starved AC-silenced flies were still capable of recovery.

Other temperature-sensing neurons involved in temperature preference behavior are cold-sensing *R11F02-Gal4*-expressing neurons^{26,64}. To determine whether *R11F02-Gal4*-expressing neurons are involved in taste-evoked warm preference, we silenced *R11F02-Gal4*-expressing neurons using *uas-Kir*. We found that the flies showed a significant difference in Tp between fed and starved conditions, but the flies did not show a warm preference upon sucralose refeeding (**Fig. 3B**, gray and blue bars, statistics shown as green and red stars, Table S1). As controls, *TrpA1^{SH}-Gal4/+*, *R11F02-Gal4/+* and *uas-Kir/+* flies showed normal starvation response and taste-evoked warm preference (**Fig. 3C–3E**).

To further ensure the results, we used optogenetics to artificially excite warm and cold neurons with *TrpA1^{SH}-Gal4* and *R11F02-Gal4*, respectively, by red light pulses for 10 min. We compared the Tp of starved flies with and without ATR under red light. Tp of starved flies with ATR was significantly increased (**Fig. 3F** and **3G**, yellow bars, statistics shown as green and red stars, Table S1) compared to those without ATR (**Fig. 3F** and **3G**, gray bars, Table S1). Therefore, these data indicate that ACs and *R11F02-Gal4*-expressing neurons are required for taste-evoked warm preference.

Next, we asked whether temperature-sensitive neurons contribute to the nutrient-induced warm preference. We used the warm- or cold neuron silenced flies (*TrpA1^{SH}-Gal4* or *R11F02-Gal4>uas-Kir*) and found that all these starved flies did not increase Tp after 10 min glucose intake (**Fig. 3A** and **3B**, green bars, Table S1), but increased Tp after 10 min refeeding with fly food containing carbohydrate, fat, and protein (**Fig. 3A** and **3B**, orange bars, Table S1). Notably, AC-silenced flies increased Tp after 1 hr glucose intake (**Fig. 3A**, green bars, Table S1). All control flies showed normal responses to both 10 min glucose refeeding and fly food intake (**Fig. 3C–3E**, green bars, Table S1). The data suggest that temperature-sensing neurons are required for warm preference in carbohydrate refeeding, but not in other foods such as fat or protein (see Discussion).

Olfaction is possibly involved in a warm preference for hungry flies

We also investigated the potential effects of olfaction. We used mutants of the odorant receptor co-receptor, *Orco* (*Orco¹*), which has an olfactory defect⁶⁶. We found that the flies showed a significant difference in Tp between fed and starved conditions, but the flies did not show a warm preference upon sucralose refeeding (**Fig. S4A** and **S4C**, gray and blue bars, statistics shown as green and red stars, Table S1). We found that all of these starved flies increased Tp after 10 min of glucose or fly food intake and showed a full recovery after 1 hr of glucose intake (**Fig. S4B** and **S4C**, orange and green bars, statistics shown as green and red stars, Table S1). The data suggest that olfaction may be involved in the warm preference in sucralose refeeding.

Internal state influences taste-evoked warm preference in hungry flies

Internal state strongly influences motivation to feed. However, how internal state influences starving animals to exhibit a food response remains unclear. Hunger represents the food-deficient state in the body, which induces the release of hunger signals such as neuropeptide Y (NPY). NPY promotes foraging and feeding behavior in mammals and flies⁶⁷. While intracerebroventricular injection of NPY induces the cephalic phase response (CPR)⁶⁸, injection of NPY antagonists

suppresses CPR in dogs, suggesting that NPY is a regulator of CPR in mammals ⁶⁹. Therefore, we first focused on neuropeptide F (NPF) and small neuropeptide F (sNPF), which are the *Drosophila* homologue and orthologue of mammalian NPY, respectively ⁶⁷, and asked whether they are involved in taste-evoked warm preference. In *NPF* mutant (*NPF*^{-/-}) or *sNPF* hypomorph (*sNPF hypo*) mutant, we found that the Tp of fed and starved flies were significantly different, showing a normal starved response (Fig. 4A and 4B: white and gray bars, statistics shown as green and red stars, Table S1). However, they failed to show a taste-evoked warm preference after 10 min of sucralose refeeding (Fig. 4A and 4B: blue bars, statistics shown as green and red stars, Table S1). Thus, NPF and sNPF are required for taste-evoked warm preference after sucralose refeeding.

We next asked whether NPF and sNPF are involved in the nutrient-induced warm preference during glucose refeeding. We found that starved *NPF*^{-/-} mutants significantly increased Tp after 1 hr of glucose refeeding (Fig. 4A, dark green bar, statistics shown as green and red stars, Table S1). We also found that starved *sNPF hypo* mutants significantly increased Tp after 10 min and 1 hr of glucose refeeding (Fig. 4B, green and dark green bars, statistics shown as green and red stars, Table S1). In addition, both *NPF*^{-/-} and *sNPF hypo* mutants increased Tp after 10 min of fly food refeeding (Fig. 4A and 4B, orange bars, statistics shown as green and red stars, Table S1). However, when these flies were fed both glucose and fly food, they did not reach the same Tp as the fed flies. These data suggest that NPF and sNPF also play a role in the modulation of nutrient-induced warm preference.

Factors involving hunger regulate taste-evoked warm preference

Based on the above results, we hypothesized that hunger signals might be involved in taste-evoked warm preference. To test this hypothesis, we focused on several factors involved in the hunger state. The major hunger signals ²⁴, diuretic hormone 44 (DH44), and adipokinetic hormone (AKH) are the mammalian corticotropin-releasing hormone homologue ⁷⁰ and the functional glucagon homologue, respectively ^{71,72}. We found that DH44- or AKH-expressing neuron silenced flies failed to show a taste-evoked warm preference after sucralose refeeding. However, they were able to increase Tp after 10 min glucose or fly food refeeding (Fig. 4D and 4F, green and orange bars, statistics shown as green and red stars, Table S1), suggesting a normal nutrient-induced warm preference. The control flies (*Dh44-Gal4/+*, *Akh-Gal4/+*, and *UAS-Kir/+*) showed a normal warm preference to sucralose (Fig. 4C, 4E and 4J, blue bars, statistics shown as green and red stars, Table S1), glucose (Fig. 4C, 4E and 4J, green bars, statistics shown as green and red stars, Table S1), and normal fly food refeeding (Fig. 4C, 4E and 4J, orange bars, statistics shown as green and red stars, Table S1). The data suggest that DH44 or AKH neurons are required for taste-evoked warm preference, but not for nutrient-induced warm preference.

Insulin-like peptide 6 (Ilp6) is a homologue of mammalian insulin-like growth factor 1 (IGF1), and *ilp6* mRNA expression is increased in starved flies ⁷³⁻⁷⁵. Because Ilp6 is important for the hunger-driven lower Tp ²⁶, the *ilp6* mutants did not show a significant difference in Tp between fed and starved conditions for only one overnight starvation. Therefore, we first extended the starvation time to three overnights. We found that the *ilp6* loss-of-function (*ilp6 LOF*) mutant failed to show a taste-evoked warm preference after sucralose refeeding (Fig. 4G, blue bars, statistics shown as green and red stars, Table S1), but did show a nutrient-induced warm preference after glucose refeeding (Fig. 4G, green bars, statistics shown as green and red stars, Table S1). These data suggest that Ilp6 is required for taste-evoked warm preference but not for nutrient-induced warm preference after glucose or fly food refeeding.

Unpaired3 (Upd3) is a *Drosophila* cytokine that is upregulated under nutritional stress ⁷⁶. We found that the *upd3* mutants failed to show a taste-evoked warm preference after sucralose refeeding (Fig. 4H, blue bar, statistics shown as green and red stars, Table S1), but showed a nutrient-induced warm preference after glucose or normal food refeeding (Fig. 4H, green and

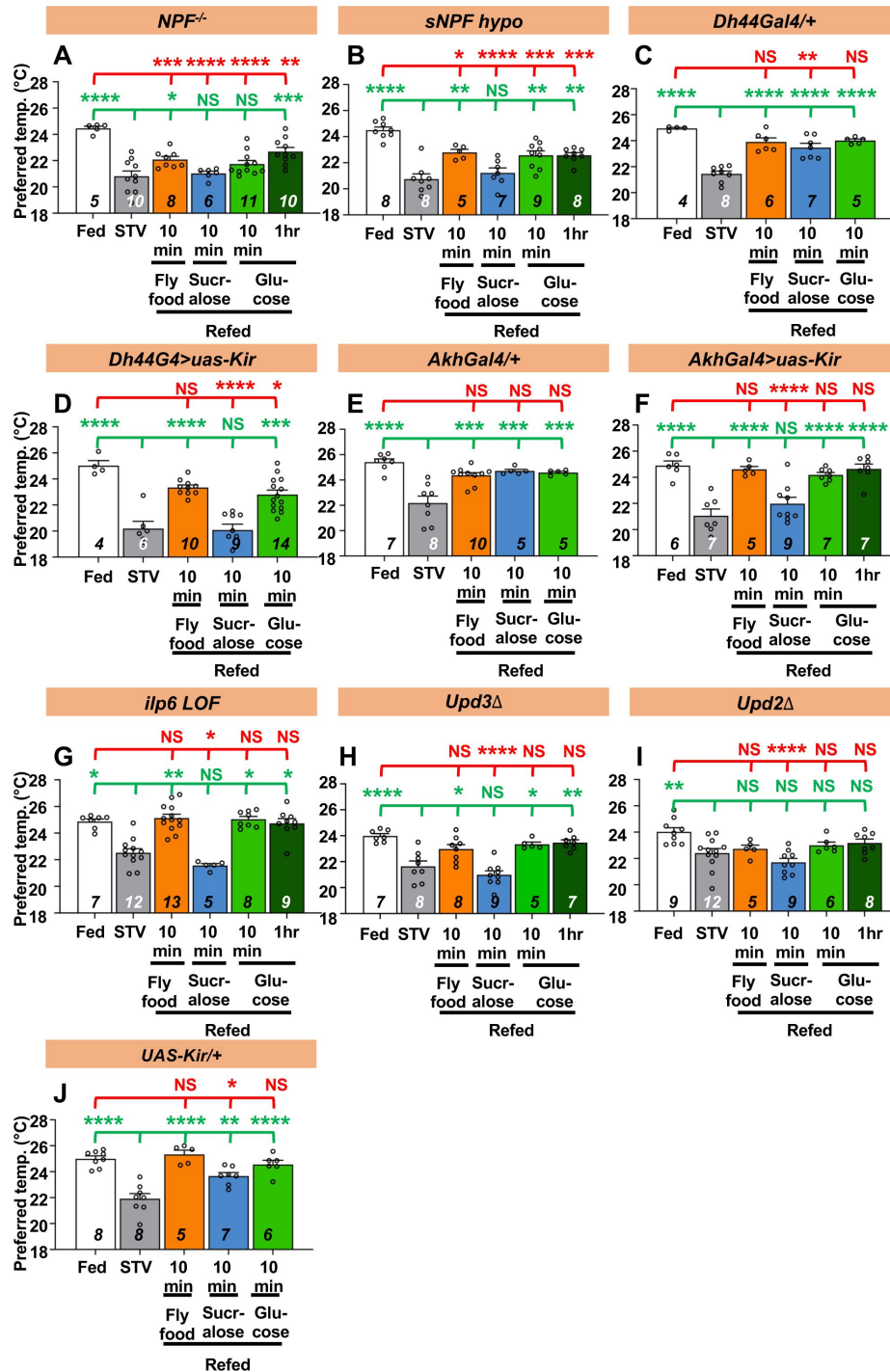


Figure 4.

Hunger signals are involved in taste-evoked warm preference.

Comparison of preferred temperature (Tp) of flies between fed (white bar), starved (STV; gray bar), and refeed states (orange, blue, green, or dark green). Flies were starved for 2 overnights (ON) except for *ilp6* mutant flies (3ON). Starved flies were refeed with fly food (orange bar), sucralose (blue bar), or glucose (green bar) for 10 min. These behavioral experiments were conducted on ZT4-7. Dots on each bar indicate individual Tp in the assays. Numbers in italics indicate the number of trials. Shapiro-Wilk test was performed for normality test. One-way ANOVA was used for statistical analysis. Red or green stars indicate Tukey's post hoc test comparing between each experiment to the fed (red) or starved (green) condition, respectively. All data presented are means with SEM. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. NS indicates not significant.

orange bars, statistics shown as green and red stars, Table S1). These data suggest that Upd3 is required for taste-evoked warm preference, but not for nutrient-induced warm preference after glucose or normal food refeeding.

We also examined the role of the satiety factor Unpaired2 (Upd2), a functional leptin homologue in flies ⁷⁷, in taste-evoked warm preference. The *upd2* mutants failed to show a warm preference after refeeding of sucralose, glucose or fly food (Fig. 4I, blue, green and orange bars, statistics shown as green and red stars, Table S1). Thus, our data suggest that factors involved in the hunger state are required for taste-evoked warm preference.

Flies show a taste-evoked warm preference at all times of the day

Animals anticipate feeding schedules at a time of day that is tightly controlled by the circadian clock ⁴¹. Flies show a rhythmic feeding pattern: one peak in the morning ⁵⁰ or two peaks in the morning and evening ⁷⁸. Because food cues induce a warm preference, we wondered whether feeding rhythm and taste-evoked warm preference are coordinated. If so, they should show a parallel phenotype.

Since flies exhibit one of the circadian outputs, the temperature preference rhythm (TPR) ²¹, Tp gradually increases during the day and peaks in the evening. First, we tested starvation responses at Zeitgeber time (ZT)1-3, 4-6, 7-9, and 10-12 under light and dark (LD) conditions, with flies being offered only water for 24 h prior to the experiments at each time point. We found that both *w¹¹¹⁸* and *yellow¹ white¹ (y¹w¹)* flies had higher Tp in the fed state and lower Tp in the starved state at all times of the day (Fig. 5A and 5B; black and gray lines; Table S1). Next, we refed sucralose to starved flies at all time points tested and examined taste-evoked warm preference. While starved *y¹w¹* flies showed a taste-evoked warm preference at all time points (Fig. 5B1, gray and blue lines, Table S1), starved *w¹¹¹⁸* flies showed a significant taste-evoked warm preference at ZT4-6 and 10-12 (Fig. 5A1, gray and blue lines, Table S1).

Because starved *w¹¹¹⁸* flies showed an advanced phase shift of TPR with a peak at ZT7-9 (Fig. 5A, gray), it is likely that the highest Tp simply masks the taste-evoked warm preference at ZT7-9. We also focused on nutrient- (carbohydrate-) induced warm preference. Starved *w¹¹¹⁸* and *y¹w¹* flies successfully increased Tp after 10 min of glucose refeeding (Fig. 5A2 and B2, green line, Table S1). Glucose refeeding for 1 hr resulted in Tp similar to that of fed *w¹¹¹⁸* flies (Fig. 5A3, dark green line). Because the feeding rhythm peaks in the morning or morning/evening ^{50,78}, our data suggest that the feeding rhythm and taste-evoked warm preference do not occur in parallel.

Circadian clock genes are required for taste-evoked warm preference, but not for nutrient-induced warm preference

We asked whether the circadian clock is involved in taste-evoked warm preference. We used clock gene null mutants, *period⁰¹ (per⁰¹)* and *timeless⁰¹ (tim⁰¹)*. Although they showed significant starvation responses (Fig. 5C and 5D, black and gray lines, Table S1), neither starved *per⁰¹* nor *tim⁰¹* mutants could show taste-evoked warm preference upon sucralose refeeding (Fig. 5C1 and 5D1, blue lines, Table S1). Nevertheless, they fully recovered upon glucose refeeding in LD at any time of day (Fig. 5C2 and 5D2, green lines, Table S1). Therefore, our data suggest that clock genes are required for taste-evoked warm preference, but not for nutrient-induced warm preference.

However, starved *per⁰¹* and *tim⁰¹* mutants may eat sucralose less frequently than glucose, which could result in a failure to show a taste-evoked warm preference. Therefore, we examined how often starved *per⁰¹* and *tim⁰¹* mutants touched glucose, sucralose, or water during the 30 min using FLIC assays ⁵⁰ (Fig. S5A-S5C). Interestingly, starved *per⁰¹* and *tim⁰¹* mutants touched water significantly more often than glucose or sucralose (Fig. S5B and S5C, Table S1).

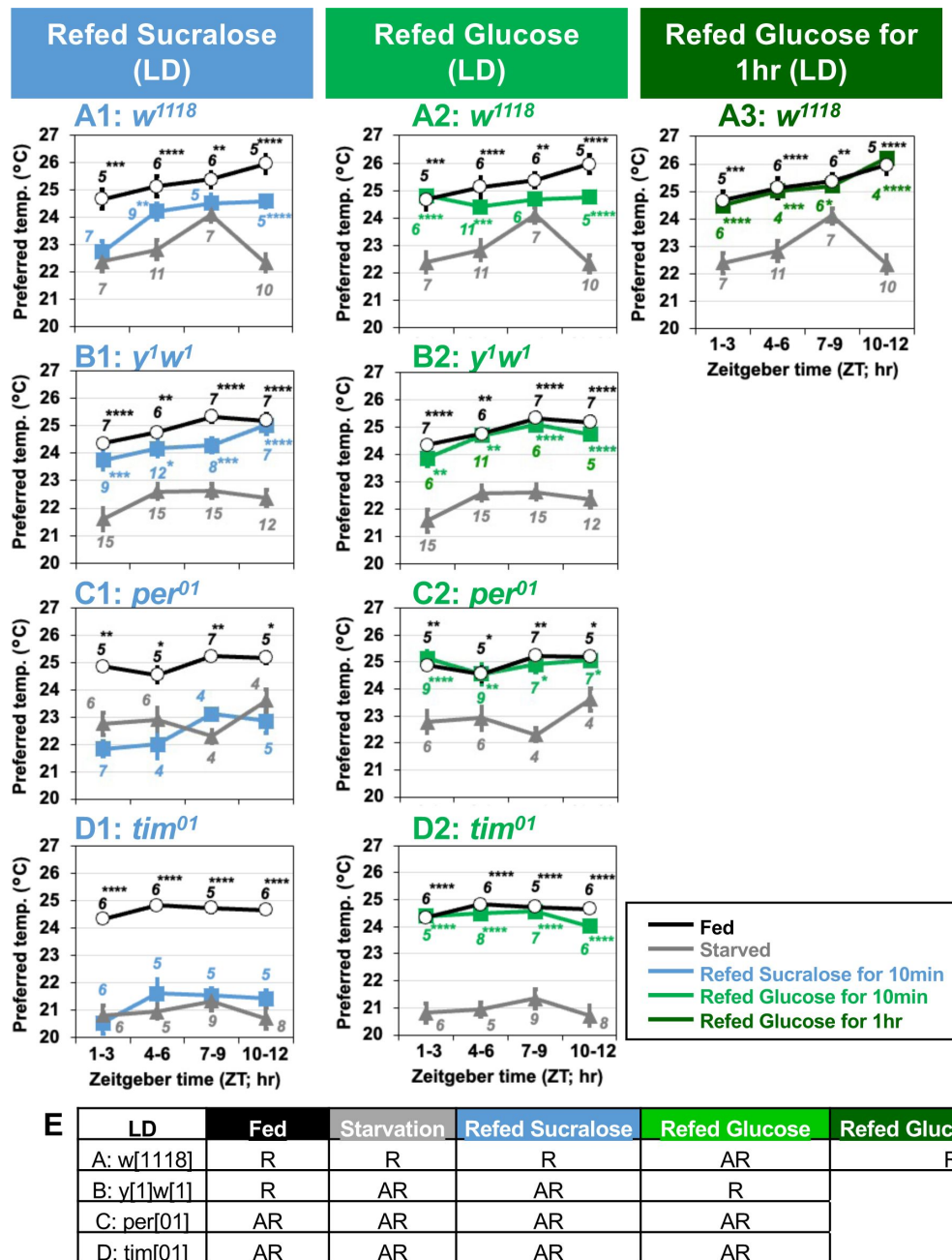


Figure 5.

Clock genes are involved in taste-evoked warm preference.

(A-D) Comparison of preferred temperature (Tp) of flies between fed (white circles), starved (STV; gray triangles), and refeed (blue, green, or dark green squares) states. Flies were starved for 24 hrs. Starved flies were refeed with sucralose (blue squares, A1-D1) or glucose for 10 min (green squares, A2-D2) or 1 hr (dark green squares, A3). After the 10 min or 1 hr refeeding, the temperature preference behavior assays were performed immediately at ZT1-3, ZT4-6, ZT7-9, and ZT10-12 in LD. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Stars indicate Tukey's post hoc test or Dunn's test compared between each experiment and the starved condition at the same time point. All data presented are means with SEM. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. (E) Comparison of Tp during the daytime in each feeding state. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis between ZT1-3 and ZT7-9 or ZT10-12, respectively. R and AR indicate rhythmic and arrhythmic, respectively, during the daytime.

Although starved *tim*⁰¹ flies touched glucose slightly more than sucralose for only 10 min, this phenotype is not consistent with *per*⁰¹ and *w*¹¹¹⁸ flies. However, these mutants still showed a similar Tp pattern for sucralose and glucose refeeding (Fig. 5C and 5D). The results suggest that although the *tim*⁰¹ flies can eat sufficient amount of sucralose over glucose, their food intake does not affect the Tp behavioral phenotype. Thus, we conclude that in *per*⁰¹ and *w*¹¹¹⁸ flies, the differential response between taste-evoked and nutrient-induced warm preferences is not due to feeding rate.

Discussion

When animals are hungry, sensory detection of food (sight, smell, or chewing) initiates digestion even before the food enters the stomach. The food-evoked responses are also observed in thermogenesis, heart rate, and respiratory rate in mammals^{14,15,79}. These responses are referred to as the cephalic phase response (CPR) and contribute to the physiological regulation of digestion, nutrient homeostasis, and daily energy homeostasis¹¹. While starved flies show a cold preference, we show here that the food cue, such as the excitation of gustatory neurons, triggers a warm preference, and the nutritional value triggers an even higher warm preference. Thus, when flies exit the starvation state, they use a two-step approach to recovery, taste-evoked and nutrient-induced warm preferences. The taste-evoked warm preference in *Drosophila* may be a physiological response potentially equivalent to CPR in mammals. Furthermore, we found that internal needs, controlled by hunger signals and circadian clock genes, influence taste-evoked warm preference. Thus, we propose that the taste-evoked response plays an important role in recovery and represents another layer of regulation of energy homeostasis.

Tp is determined by the taste cue

Starved flies increase Tp in response to a nutrient-free taste cue (Fig. 1C and 1F), resulting in a taste-evoked warm preference. We showed that silencing of ACs or cold neurons caused a loss of taste-evoked warm preference (Fig. 3A–3E), and that excitation of ACs or cold neurons induced a taste-evoked warm preference (Fig. 3F and 3G). The data suggest that both warm and cold neurons are important for taste-evoked warm preference: while ACs are required for the hunger-driven lower Tp²⁶, both ACs and cold neurons are likely to be important for this taste-evoked warm preference.

The hunger-driven lower Tp is a slower response because starvation gradually lowers their Tp²⁶. In contrast, the taste-evoked warm preference is a rapid response. Once ACs and cold neurons are directly or indirectly activated, starved flies quickly move to a warmer area. This is interesting because even when ACs and cold neurons are activated by warm and cold, respectively, the activation of these neurons causes a warm preference. Given that the sensory detection of food (sight, smell, or chewing food) triggers CPR, the activation of these sensory neurons may induce CPR. Tp in sucralose-refed hungry flies is between that of fed and starved flies (Figs. 2 and 3), making it difficult to detect the smaller temperature differences using the calcium imaging experiments. Therefore, we speculate that both ACs and cold neurons may facilitate rapid recovery from starvation so that flies can quickly return to their preferred temperature - body temperature - to a normal state.

Internal state influences taste-evoked warm preference

We show that mutants of genes involved in hunger and the circadian clock fail to show taste-evoked warm preference, suggesting that hunger and clock genes are important for taste-evoked warm preference. At a certain time of day, animals are hungry for food^{50,78}. Thus, the hungry state acts as a gatekeeper, opening the gate of the circuits when hungry flies detect the food information that leads to taste-evoked warm preference (Figs. 4 and S6, blue arrow). While

most of the hunger signals we focused on are important for taste-evoked warm preference, some hunger signals are also required for both taste-evoked and nutrient-induced warm preferences (Fig. 4). Notably, sensory signals contribute to both taste-evoked and nutrient-induced warm preferences (Figs. 2, 3 and 56, blue and green arrows). Thus, taste-evoked warm preference and nutrient-induced warm preference differ at the internal state level, but not at the sensory level. This idea is analogous to appetitive memory formation. Sweet taste and nutrients regulate the different layers of the memory formation process. Recent evidence suggests that the rewarding process can be subdivided; the sweet taste is for short-term memory and the nutrient is for long-term memory^{80–82}. The data suggest that the taste-evoked response functions differently from the nutrient-induced response. Therefore, taste sensation is not just the precursor to nutrient sensing/absorption, but plays an essential role in the rapid initiation of a taste-evoked behavior that would help the animal survive.

How do hunger signals or clock genes contribute to taste-evoked warm preference?

The hunger signaling hormones/peptides studied in this project are important for taste modulation. For example, mammalian NPY and its *Drosophila* homolog NPF modulate the output of taste signals^{49,83,84}. The AKH receptor is expressed in a subset of gustatory neurons that may modulate taste information for carbohydrate metabolism⁸⁵. Therefore, the hunger signals are likely to modulate the downstream of the sensory neurons, which may result in a taste-evoked warm preference.

Circadian clock genes control and coordinate the expression of many clock-controlled genes in the body^{86–89}. Therefore, we expect that the absence of clock genes will disrupt the molecular and neural networks of homeostasis, including metabolism, that are essential for animal life. For example, taste neurons express clock genes, and impaired clock function in taste neurons disrupts daily rhythms in feeding behavior⁹⁰. Temperature-sensing neurons transmit hot or cold temperature information to central clock neurons^{91–95}. Therefore, the disrupted central clock in clock mutants may respond imprecisely to temperature signals. There are many possible reasons why the lack of clock gene expression in the brain is likely to cause abnormal taste-evoked warm preference.

In addition, hunger signals may contribute to the regulation of circadian output. DH44 is located in the dorsomedial region of the fly brain, the pars intercerebralis (PI), and DH44-expressing neurons play a role in the output pathway of the central clock^{96,97}. Insulin-producing cells (IPCs) are also located in addition to DH44-expressing neurons^{98–101}. IPCs receive a variety of information, including circadian^{97,102} and metabolic signals^{73–75,77}, and then transduce the signals downstream to release IIPs. Both Upd2 and Ilp6, which are expressed in the fat body respond to metabolic states and remotely regulate IIP expression^{73–75,77}. Insect fat body is analogous to the fat tissues and liver in the vertebrates^{103,104}. Therefore, each hunger signal may have its specific function for taste-evoked warm preference. Further studies are needed to describe the entire process.

Taste-evoked warm preference may be CPR in flies

The introduction of food into the body disrupts the internal milieu, so CPR is a necessary process that helps animals prepare for digestion. Specifically, in mammals, taste leads to an immediate increase in body temperature and metabolic rate. Starvation results in lower body temperatures, and chewing food, even before it enters the stomach, triggers a rapid increase in heat production, demonstrating CPR in thermogenesis^{14,15}.

Starved flies have a lower T_p ²⁶. Because *Drosophila* is a small ectotherm, the lower T_p indicates a lower body temperature^{19,20}. Even when the flies do not receive food, the sweet taste and the excitation of sweet neurons induce starved flies to show a warm preference, which eventually

leads to a warmer body temperature. In fact, CPR is known to be influenced by smell as well in mammals ⁹. We have shown in flies that olfactory mutants fail to show a warm preference when refed sucralose (**Fig. S4**). Starvation leads to lower body temperatures, and food cues, including taste and odor, rapidly induce a rise in body temperature before food enters the body. Thus, the taste-evoked warm preference in *Drosophila* may be a physiological response equivalent to one of the CPRs observed in mammals.

CPR is essential because both starved mammals and starved flies must rapidly regulate their body temperature to survive

As soon as starved flies taste food, the sensory signals trigger CPR. They can move to a warmer place to prepare to raise their body temperature (**Fig. S6**, blue arrows). CPR may allow flies to choose a more hospitable place to restore their physiological state and allow for a higher metabolism, and eventually move on to the next step, such as foraging and actively seeking a mate before competitors arrive. Thus, CPR may be a strategy for the fly's survival. Similarly, starvation or malnutrition in mammals leads to lower body temperatures ^{6,7,105}, and biting food triggers heat production, which is CPR ¹⁴⁻¹⁶. Thus, while starvation in both flies and mammals leads to lower body temperatures, food cues initiate CPR by increasing body temperature and nutrient intake, resulting in full recovery from starvation. Our data suggest that *Drosophila* CPR may be a physiological response equivalent to CPR observed in other animals. Thus, *Drosophila* may shed new light on the regulation of CPR and provide a deeper understanding of the relationship between CPR and metabolism.

Acknowledgements

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Contributions

F.N.H. and Y.U. designed the experiments. Y.U., E.N., T.N. and J.S. performed the temperature preference behavioral assays and data analysis. Y.U., S.H.S., S.S.U. and J.C. performed the feeding experiments and the data analysis. F.N.H. and Y.U. wrote the manuscript.

Declaration of Interest

The authors declare no competing interests

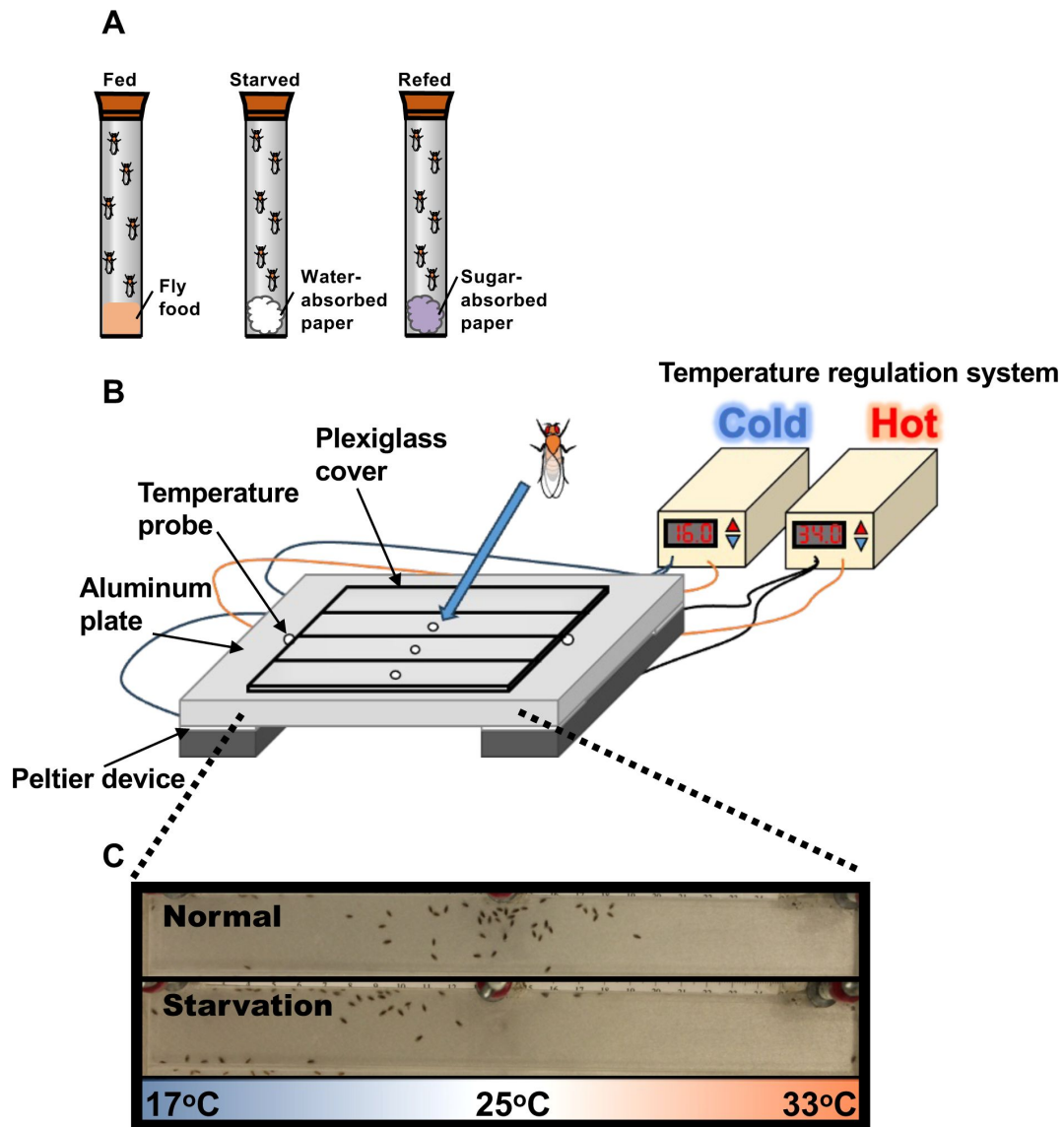


Figure S1.

***Drosophila* temperature preference behavior assay (Related to Fig. 1)**

(A) Three different feeding conditions, fed, starved and refed, were used in this study.

(B) A temperature gradient of 17-33°C in air is established in the chamber between the aluminum metal plate and the Plexiglas cover. Flies are applied through the hole into the chamber. Once the flies are applied, they spread out in the chamber and then gradually accumulate at the specific temperature ranges within 30 min, which is called the preferred temperature (Tp).

(C) One of the representative results of Tp experiments (top, normal (fed flies); bottom, starvation (starved flies)). Since their body temperature is close to the temperature of their surrounding microenvironment, their body temperature is determined by measuring their Tp.

Figure S2.

The distribution of temperature preference in each experimental condition (Related to Fig. 1 and 2)

Comparison of the distribution of temperature preference in each feeding condition in w^{1118} control flies (A) and the $Gr64fGal4>CsChrimson$ flies (B). Heat maps were generated using the Conditional formatting tool in Microsoft Excel. The averaged percentages of flies settling on the apparatus within each one-degree temperature interval were used to create the heat maps. Each parameter to draw them using Conditional formatting tool is as follows: Minimum Value: 0, Midpoint Value: 15%, and Maximum Value: 60% for w^{1118} . Minimum Value: 0, Midpoint Value: 10%, and Maximum Value: 45% for $Gr64fGal4>CsChrimson$.

A: w^{1118}

Feeding condition	Temperature gradient (°C)															
	17-18	18-19	19-20	20-21	21-22	22-23	23-24	24-25	25-26	26-27	27-28	28-29	29-30	30-31	31-32	32-33
Fed	0%	0%	0%	3%	1%	4%	3%	19%	51%	17%	1%	0%	0%	0%	0%	1%
STV	15%	8%	7%	13%	12%	13%	11%	11%	4%	4%	3%	0%	0%	0%	0%	0%
Refed fly food for 5min	6%	6%	6%	5%	10%	8%	12%	18%	15%	8%	4%	1%	1%	0%	1%	0%
Refed fly food for 10min	1%	2%	2%	2%	3%	3%	10%	33%	31%	8%	2%	1%	1%	1%	0%	0%
Refed fly food for 30min	0%	0%	0%	0%	2%	5%	13%	41%	25%	11%	0%	1%	0%	1%	0%	0%
Refed fly food for 1hr	0%	0%	0%	2%	1%	6%	7%	25%	31%	20%	4%	1%	0%	1%	1%	0%
Refed Sucralose for 10min	2%	0%	1%	9%	19%	17%	19%	20%	7%	3%	2%	1%	0%	0%	0%	0%
Refed Sucralose for 1hr	5%	5%	8%	6%	9%	7%	23%	17%	11%	5%	1%	1%	1%	0%	0%	0%
Refed Glucose for 5min	3%	1%	2%	7%	8%	12%	16%	27%	14%	5%	3%	1%	0%	1%	0%	0%
Refed Glucose for 10min	1%	0%	3%	2%	5%	10%	16%	23%	26%	9%	1%	2%	0%	1%	0%	0%
Refed Glucose for 1hr	1%	0%	1%	1%	1%	4%	9%	23%	43%	14%	3%	1%	1%	0%	0%	0%

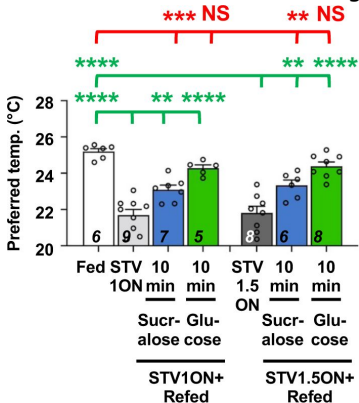
B: $Gr64fGal4>uas-CsChrimson$

Feeding condition	Temperature gradient (°C)															
	17-18	18-19	19-20	20-21	21-22	22-23	23-24	24-25	25-26	26-27	27-28	28-29	29-30	30-31	31-32	32-33
Fed	0%	2%	1%	1%	1%	5%	8%	10%	43%	25%	5%	0%	0%	0%	0%	0%
STV without ATR and red light exposure for 10min	7%	6%	5%	11%	5%	10%	15%	13%	19%	8%	0%	0%	0%	0%	0%	0%
STV with ATR and red light exposure for 10min	2%	4%	3%	4%	4%	10%	16%	25%	22%	6%	2%	0%	0%	1%	0%	0%

Figure S3.

Duration of starvation is unlikely to affect the ability to recover (Related to Fig. 3)

Comparison of preferred temperature (T_p) of w^{1118} control flies between fed (white bar), starved (STV; gray bar), and refed conditions (blue or green bar). Flies were starved for overnight (STV1ON; gray bar) or 1.5 ON (STV1.5ON; dark gray bar), i.e. starved for 18-21 hrs or 26-29 hrs, respectively. Starved flies were refed with sucralose for 10 min (blue bar) or glucose for 10 min (green bar). These behavioral experiments were performed on ZT4-7. Dots on each bar indicate individual T_p in the assays. Numbers in italics indicate the number of trials. Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Red or green stars indicate Tukey's post hoc test or Dunn's test comparing differences between experimental and fed or starved conditions, respectively. All data shown are means with sem. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. **** $p < 0.0001$. NS indicates no significance. The same data from Fed, STV1ON, STV1ON+refed sucralose for 10 min and STV1ON+refed sucralose for 10 min in Fig. 1B-F are used in Fig. S3.



Orco1

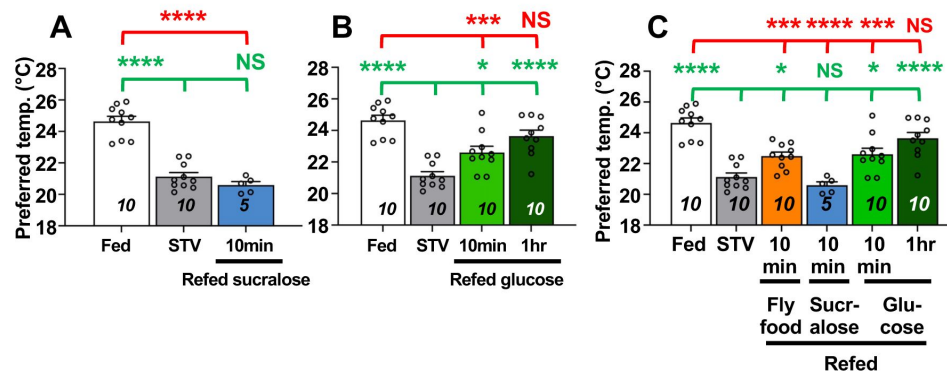


Figure S4.

Orco is involved in the taste-evoked warm preference.

Comparison of the preferred temperature (Tp) of *Orco1*¹⁰⁰⁰ olfaction-deficient mutant flies between fed (white bar), starved (STV; gray bar), and refed (blue or green bar). *Orco1*¹⁰⁰⁰ flies were starved for 2 overnights (ON). Starved flies were refed with sucralose for 10 min (blue bar; A) or glucose for 10 min (green bar; B). Pooled data of both sucralose- and glucose-refeeding bioassay data are shown in C. The Shapiro-Wilk test was performed to test for normality. One-way ANOVA or Kruskal-Wallis test was used for statistical analysis. Red or green stars indicate Tukey's post hoc test or Dunn's test compared between each experiment and the fed (red) or starved (green) condition, respectively.

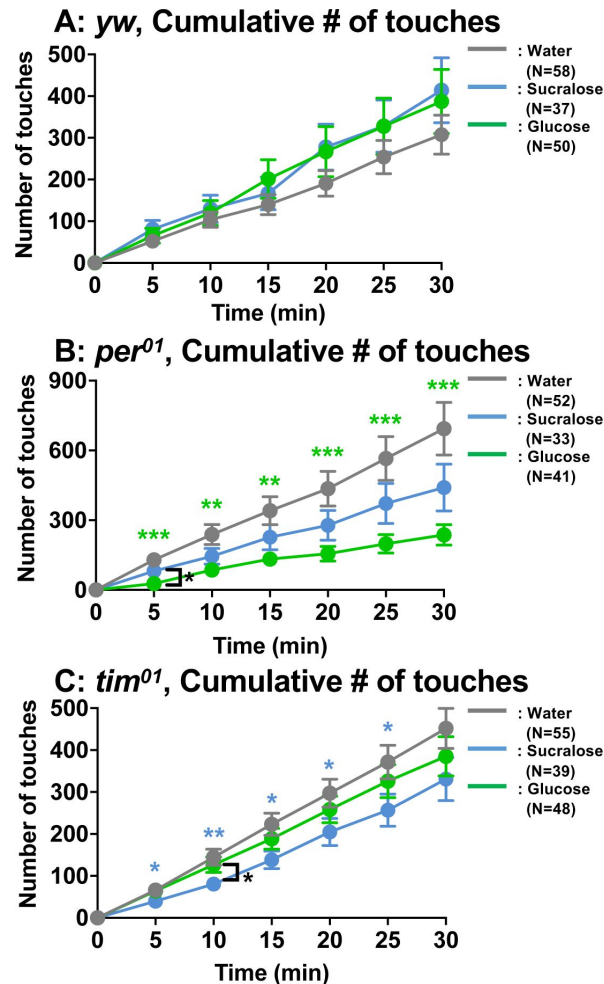


Figure S5.

Number of touches to water, sucralose and glucose using *yw*, *per*⁰¹ and *tim*⁰¹ flies. (Related to Figures 1 and 5)

Feeding assay: The number of touches to water (gray), 2.8 mM (equivalent to 5%) glucose (green), or 2.8 mM sucralose solution (blue) was examined using *yw*, *per*⁰¹, or *tim*⁰¹ flies starved for 24 hrs. Cumulative number of touches for 0-30 min was plotted. Two-way ANOVA was used for multiple comparisons. Blue and green stars show Fisher's LSD post hoc test comparing sucralose (blue stars) or glucose (green stars) solution feeding to water drinking. Black stars show the statistical difference between sucralose and glucose refeeding. All data presented are means with SEM. *p < 0.05. *F*p < 0.01. ***p < 0.001. ****p < 0.0001.

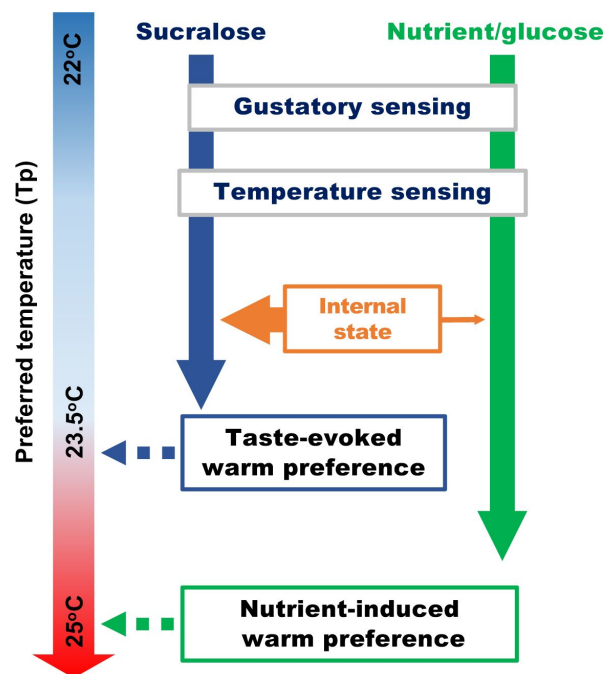


Figure S6.

Model

A schematic diagram of the recovery of preferred temperature (T_p) by taste-evoked warm preference (blue arrows) and nutrient-induced warm preference (green arrows).

Materials and Methods

All flies were reared under 12 hr-light/12 hr-dark (LD) cycles at 25°C and 60-70% humidity in an incubator (DRoS33SD, Powers Scientific Inc.) with an electric timer (light on: 8am; light off: 8pm). The light intensity was 1000-1400 lux. All flies were reared on custom fly food recipe, with the following composition per 1 L of food: 6.0 g sucrose, 7.3 g agar, 44.6 g cornmeal, 22.3 g yeast and 16.3 mL molasses, as previously described²⁶. *white*¹¹¹⁸ (*w*¹¹¹⁸) and *yellow*¹ *white*¹ (*y*¹ *w*¹) flies were used as control flies. *EP5A*; *Gr64a*¹ (*Gr5a*^{-/-}; *Gr64a*^{-/-}) was kindly provided by Dr. Anupama Dahanukar⁴⁷. *R1*; *Gr5a-LexA*; +; Δ *Gr61a*, Δ *Gr64a-f* (*Gr5a*^{-/-}; *Gr61a*^{-/-}, *Gr64a-f*^{-/-}) were kindly provided by Dr. Hubert Amrein^{51,52}. *Dh44-Gal4* was kindly provided by Dr. Greg Suh⁴⁸. *TrpA1^{SH}-gal4* was kindly provided by Dr. Paul A. Garrity⁴⁴. Other fly lines were provided by Bloomington Drosophila stock center and Vienna Drosophila Stock Center.

Temperature preference behavioral assay

The temperature preference behavior assays were examined using a temperatures gradient, set from 16-34°C and were performed for 30 min in an environmental room maintained at 25°C /60%-70% humidity, as previously described²⁶. Because starved flies showed a lower preferred temperature (*T_p*), the temperature regulation was lower than usual (Umezaki et al. 2018 Current Biology). We prepared a total of 40-50 flies (male and female flies were mixed) for fed condition experiments and 90-100 flies for overnight(s) starved and refed condition experiments for one trial. Flies were never reused in subsequent trials. In the starved and refed experiments, we prepared twice the number of flies needed for a trial because almost half of them died from starvation stress. Others climbed on the wall and ceiling (starved flies are usually hyperactive¹⁰⁶), even though we applied slippery coating chemicals (PTFE; Cat# 665800, Sigma or byFormica PTFE Plus, <https://byformica.com/products/fluon-plus-ptfe-escape-prevention-coating>) to the Plexiglass covers.

Behavioral assays were performed for 30 min at Zeitgeber time (ZT) 4-7 (light on and light off are defined as ZT0 and ZT12, respectively), and starvation was initiated at ZT9-10 (starved for 1, 2 or 3 O/N are 18-21 hr, 42-45 hr or 66-69 hr, respectively). As for STV1.5 for Fig. S3, starvation was initiated at ZT1-2 (26-29 hr). For the starvation assays, the collected flies were maintained on our fly food for at least one day and then transferred to plastic vials containing 3 mL of distilled water, which was absorbed by a Kimwipe paper. For refeeding experiments, starved flies were transferred to plastic vials containing 2 mL of 2.8 mM sugar solution (sucralose water/glucose water). Sugar solution is absorbed by half the size of a Kimwipe. The details of the starvation period are described in the following section (see “Starvation Condition”).

After the 30-min behavioral assay, the number of flies whose bodies were completely located on the aluminum plate was counted. Flies whose bodies were partially or completely located on the walls of the Plexiglass cover were not included in the data analysis. The percentage of flies within each one-degree temperature interval on the apparatus was calculated by dividing the number of flies within each one-degree interval by the total number of flies on the apparatus. The location of each one-degree interval was determined by measuring the temperature at 6 different points on the bottom of the apparatus. Data points were plotted as the percentage of flies within a one-degree temperature interval. The weighted mean of *T_p* was calculated by summing the product of the percentage of flies within a one-degree temperature interval and the corresponding temperature (e.g., fractional number of flies x 17.5°C + fractional number of flies x 18.5°C + fractional number of flies x 32.5°C). If the SEM of the averaged *T_p* was not < 0.3 after the five trials, additional trials were performed approximately 10 times until the SEM was < 0.3.

Microsoft Excel (Home tab>Conditional formatting tool>3-color scales and data bars) was used to create heat maps to show the distribution of flies in each experimental condition. The averaged percentages of flies that settled on the apparatus within each one-degree temperature interval were used to create the heat maps. Each scale value is as follows; minimum value: 0, midpoint value: 15%, and maximum value: 60% for *w¹¹¹⁸*. Minimum value: 0, middle value: 10%, and maximum value: 45% for *Gr64fGal4>CsChrimson*.

Starvation conditions and recovery

Most of the flies were starved for 2 overnights (O/N). Because some flies (e.g. *ilp6* mutants) show starvation resistance and seem to be still healthy even after 2 O/N of starvation. We had to starve them for 3 O/N to show a significant difference in Tp between fed and starved flies. On the other hand, some flies (e.g., *w¹¹¹⁸* flies) are very sick after 3 O/N of starvation, in which case we only had to starve them for 1 day. Therefore, the starvation conditions we used for this manuscript are from 1 to 3 O/N.

First, we confirmed the starvation period by focusing on Tp which resulted in a statistically significant Tp difference between fed and starved flies; as mentioned above, flies prefer lower temperatures when starvation is prolonged ²⁶. Therefore, when Tp was not statistically different between fed and starved flies, we extended the starvation period from 1 to 3 O/N. Importantly, we show in **Fig. S3** that the duration of starvation does not affect the recovery effect. Furthermore, *w¹¹¹⁸* flies cannot survive 42-49 or 66-69 hours of starvation.

Temperature preference rhythm (TPR) assay

For the TPR assays, we performed temperature preference behavior assays in different time windows during the daytime (zeitgeber time (ZT) or circadian time (CT) 1-3, 4-6, 7-9, and 10-12) as described previously ^{21,107}. Because starvation duration directly affects flies' Tp ²⁶, starvation was initiated at each time window to adjust the starvation duration at each time point, which means flies were starved for 24 hrs or 48 hrs but not 1 or 2 O/N. Each behavioral assay was not examined during these time periods (ZT or CT 0-1 and 11.5-12) because of large phenotype variation around light on and light off.

Furthermore, insulin levels were shown to peak at 10 min and gradually decline ¹⁰⁸. Also, how quickly the flies can consume food is unclear. These factors may influence temperature preference behavior. Therefore, to minimize these effects, we decided to test the temperature preference behavioral assay immediately after the flies had eaten the food.

Optogenetic activation

For the optogenetic activation of the target neurons for behavioral assays, the red-light-sensitive channelrhodopsin, *UAS-CsChrimson*, was crossed with each Gal4 driver. Flies were reared on fly food at 25°C and 60-70% humidity under LD cycles in an incubator (DRO33SD, Powers Scientific Inc.) with an electric timer. After the flies emerged, adult flies were collected and maintained on fly food for 1-2 days. The next day, flies were transferred to water with or without 0.8 mM all-trans retinal (ATR; #R2500, Sigma) diluted in dimethyl sulfoxide (DMSO; #472301, Sigma) for 2 O/N. To activate flies expressing *UAS-CsChrimson* crossed with Gal4 drivers, we used a 627-nm red light-emitting diode (LED) equipped with a pulsed photoillumination system (10 Hz, 0.08 mW/mm²). Flies were exposed to pulsed red light for 10 min, which corresponds to the refeeding period. This photoillumination system was used in an incubator (Sanyo Scientific, MIR-154) and followed by temperature preference behavioral assays.

The *R11F02-Gal4>uas-CsChrimson* flies do not develop into adults and die in the pupal stage. Therefore, the *Gal4/Gal80ts* system was used to restrict *uas-CsChrimson* expression. The *Gal80ts* is a temperature sensitive allele of *Gal80* that causes *Gal4* inhibition at 18°C and activation at 29°C

¹⁰⁹[\[link\]](#). The *tubGal80^{ts}; R11F02-Gal4>uas-CsChrimson* flies were reared on fly food at 18°C. Emerging adult flies were collected and kept on fly food at 29°C. The next day, flies were transferred to water (starved condition) with or without 0.8 mM ATR for 2 O/N. Starved flies with or without ATR application were exposed to pulsed red light for 10 min (equivalent to the refeeding period) and then immediately loaded into the behavioral apparatus for behavioral assays to measure their Tp. All flies were treated with ATR after they had fully developed into the adults. This means that Gal4-expressing cells were activated by red light via CsChrimson only at adult stages.

Feeding assay

To measure individual fly feeding, we used the Fly Liquid Interaction Counter (FLIC) system ⁵⁰[\[link\]](#). Groups of 1-2 day old male and female flies were starved for 24 hours starting between ZT4-5 (12-1pm). Individual flies were then loaded into FLIC monitors. Flies were acclimated to the monitors for 30 min with access to water in the feeding wells. At the start of the feeding study, water was replaced with 2.8 mM sucralose or 2.8 mM glucose solution (equivalent to 5% glucose concentration) and the number of licks (touches) was recorded for 30 min. Water, sucralose, or glucose water was administered individually in separate experiments. Assays were performed on ZT4-7. To account for the potential confounding effect of startle response when food is changed, wells where water was replaced with new water were used as a control. FLIC raw data were analyzed using the FLIC R code master (Pletcher Lab, available at https://github.com/PletcherLab/FLIC_R_Code [\[link\]](#)). Lick counts were obtained and summed in 5-min window bins, while cumulative licks were obtained by successively summing the licks in these bins.

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Editors

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Reviewer #1 (Public Review):

Summary:

This paper presents valuable findings that gustation and feeding state influence the preferred environmental temperature preference in flies. Interestingly, the authors showed that by refeeding starved animals with non-nutritive sugar sucralose, they are able to tune their preference towards a higher temperature in addition to nutrient-dependent warm preference. The authors show that temperature sensing and sweet sensing gustatory neurons (SGNs) are involved in the former but not the latter. In addition, their data indicate that peptidergic signals involved in internal state and clock genes are required for taste-dependent warm preference behavior.

The authors made an analogy of their results to the cephalic phase response (CPR) in mammals where the thought, sight and taste of food prepares the animal for the consumption of food and nutrients. The authors showed that taste triggers CPR-induced temperature preference behaviors in flies. The authors also briefly covered that the combined modalities of smell and taste induced CPR responses, showing that starved *orco* mutant flies failed to recover temperature preference after refeeding with sucralose.

The findings of this work hold promising future research prospects, for example, whether the sight of food influences temperature preference behavior in hungry flies, or whether taste, smell and sight work together or independently in promoting CPR responses.

Futhermore, these valuable behavioral results can be further investigated in flies with the advantage of being able to dissect the neural circuitry underlying CPR and nutrient homeostasis.

Strengths:

- (1) The authors convincingly showed that tasting is sufficient to drive warm temperature preference behavior in starved flies and show that it is independent of nutrient-driven warm preference.
- (2) By using the genetic manipulation of key internal sensors and genes controlling internal feeding and sleep state such as DH44 neurons and the *per* genes for eg the authors linked gustation and temperature preference behavior control to the internal state of the animal.

Weaknesses:

Most of the weaknesses of the paper have been addressed in the revision. The points mentioned below are meant to improve readability of the paper and to promote understanding of the significance of the work.

- (1) Supplementary fig 1 could replace Figure 1A. The purpose of Figure 1F is not clear to me as the comparison between the different food substances is not separately addressed anywhere in the text.

(2) The data for the orco receptor mutant could be placed in the main figures to justify the discussion emphasising CPR-like responses.

<https://doi.org/10.7554/eLife.94703.2.sa2>

Reviewer #2 (Public Review):

Animals constantly adjust behavior and physiology based on internal states. Hungry animals, desperate for food, exhibit physiological changes immediately upon sensing, smelling, or chewing food, known as the cephalic phase response (CPR), involving processes like increased saliva and gastrointestinal secretions. While starvation lowers body temperature, the mechanisms underlying how the sensation of food without nutrients induces behavioral responses remain unclear. Hunger stress induces changes in both behavior and physiological responses, which in flies (or at least in *Drosophila melanogaster*) leads to a preference for lower temperatures, analogous to the hunger-driven lower body temperature observed in mammals. In this manuscript, the authors have used *Drosophila melanogaster* to investigate the issue of whether taste cues can robustly trigger behavioral recovery of temperature preference in starving animals. The authors find that food detection triggers a warm preference in flies. Starved flies recover their temperature preference after food intake, with a distinction between partial and full recovery based on the duration of refeeding. Sucralose, an artificial sweetener, induces a warm preference, suggesting the importance of food-sensing cues. The paper compares the effects of sucralose and glucose refeeding, indicating that both taste cues and nutrients contribute to temperature preference recovery. The authors show that that sweet gustatory receptors (Grs) and sweet GRNs (Gustatory Receptor Neurons) play a crucial role in taste-evoked warm preference. Optogenetic experiments with CsChrimson support the idea that the excitation of sweet GRNs leads to a warm preference. The authors then examine the internal state's influence on taste-evoked warm preference, focusing on neuropeptide F (NPF) and small neuropeptide F (sNPF), analogous to mammalian neuropeptide Y. Mutations in NPF and sNPF result in a failure to exhibit taste-evoked warm preference, emphasizing their role in this process. However, these neuropeptides appear not to be critical for nutrient-induced warm preference, as indicated by increased temperature preference during glucose and fly food refeeding in mutant flies. The authors also explore the role of hunger-related factors in regulating taste-evoked warm preference. Hunger signals, including diuretic hormone (DH44) and adipokinetic hormone (AKH) neurons, are found to be essential for taste-evoked warm preference but not for nutrient-induced warm preference. Additionally, insulin-like peptide 6 (Ilp6) and Unpaired3 (Upd3), related to nutritional stress, are identified as crucial for taste-evoked warm preference. The investigation then extends into circadian rhythms, revealing that taste-evoked warm preference does not align with the feeding rhythm. While flies exhibit a rhythmic feeding pattern, taste-evoked warm preference occurs consistently, suggesting a lack of parallel coordination. Clock genes, crucial for circadian rhythms, are found to be necessary for taste-evoked warm preference but not for nutrient-induced warm preference.

Strengths:

A well-written and interesting study, investigating an intriguing issue. The claims, none of which to the best of my knowledge controversial, are backed by a substantial number of experiments.

Weakness:

The experimental setup used and the procedures for assessing the temperature preferences of flies is rather sparingly described. Additional details and data presentation would enhance the clarity and replicability of the study. I kindly request the authors to consider the following points: i) A schematic drawing or diagram illustrating the experimental setup for the

temperature preference assay would greatly aid readers in understanding the spatial arrangement of the apparatus, temperature points, and the positioning of flies during the assay. The drawing should also be accompanied by specific details about the setup (dimensions, material, etc). ii) It would be beneficial to include a visual representation of the distribution of flies within the temperature gradient on the apparatus. A graphical representation, such as a heatmaps or histograms, showing the percentage of flies within each one-degree temperature bin, would offer insights into the preferences and behaviors of the flies during the assay. In addition to the detailed description of the assay and data analysis, the inclusion of actual data plots, especially for key findings or representative trials, would provide readers with a more direct visualization of the experimental outcomes. These additions will not only enhance the clarity of the presented information but also provide the reader with a more comprehensive understanding of the experimental setup and results. I appreciate the authors' attention to these points and look forward to the potential inclusion of these elements in the revised manuscript.

Update: The revised manuscript now includes heatmaps showing the distribution of the flies across the temperature bins. As well as a schematic drawing of the behavioral setup.

<https://doi.org/10.7554/eLife.94703.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Main points:

- (1) We have added data for fructose in Fig. 1
- (2) We have added sta1s1cs (red stars and NS) comparing Tp between fed and refed flies.
- (3) We have modified the figure for each point to the opened small circles.
- (4) We have moved the data from Fig. S3 to Fig. 2 and 3.
- (5) We have added the schematic diagrams depicting behavioral assay in Fig. S1.
- (6) We have added heatmaps for WT and Gr64f-Gal4>UAS-CsChrimson flies in Fig. S2.
- (7) We have added Orco1 mutant data in Fig. S4.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This paper presents valuable findings that gustation and feeding state influence the preferred environmental temperature preference in flies. Interestingly, the authors showed that by refeeding starved animals with the non-nutritive sugar sucralose, they are able to tune their preference towards a higher temperature in addition to nutrient-dependent warm preference. The authors show that temperature-sensing and sweet-sensing gustatory neurons (SGNs) are involved in the former but not the latter. In addition, their data indicate that pep3dergic signals involved in internal state and clock genes are required for taste-dependent warm preference behavior.

The authors made an analogy of their results to the cephalic phase response (CPR) in mammals where the thought, sight, and taste of food prepare the animal for the

consumption of food and nutrients. They further linked this behavior to core regulatory genes and peptides controlling hunger and sleep in flies having homologues in mammals. These valuable behavioral results can be further investigated in flies with the advantage of being able to dissect the neural circuitry underlying CPR and nutrient homeostasis.

Strengths:

(1) The authors convincingly showed that tasting is sufficient to drive warm temperature preference behavior in starved flies and that it is independent of nutrient-driven warm preference.

(2) By using the genetic manipulation of key internal sensors and genes controlling internal feeding and sleep states such as DH44 neurons and the *per* genes for example, the authors linked gustation and temperature preference behavior control to the internal state of the animal.

Weaknesses:

(1) The title is somewhat misleading, as the term homeostatic temperature control linked to gustation only applies to starved flies.

We agree with the reviewer's suggestion and have changed the title to "Taste triggers a homeostatic temperature control in hungry flies".

(2) The authors used a temperature preference assay and refeeding for 5 minutes, 10 minutes, and 1 hour.

Experimentally, it makes a difference if the flies are tested immediately after 10 minutes or at the same time point as flies allowed to feed for 1 hour. Is 10 minutes enough to change the internal state in a nutrition-dependent manner? Some of the authors' data hint at it (e.g. refeeding with fly food for 10 minutes), but it might be relevant to feed for 5/10 minutes and wait for 55/50min to do the assays at comparable time points.

Thank you for your suggestions. The temperature preference behavioral test itself takes 30 minutes from the time the flies are placed in the apparatus until the final choice is made. This means that after the hungry flies have been refed for 5 minutes, they will determine their preferred temperature within 35 minutes. It has been shown that insulin levels peak at 10 minutes and gradually decline (Tsao, et al., PLoS Genetics 2023). However, it is unclear how subtle insulin levels affect behavior and how quickly the flies are able to consume food. These factors may contribute to temperature preference in flies. Therefore, to minimize "extraneous" effects, we decided to test the behavioral assay immediately after they had eaten the food. We have noted in the material and method section that why we chose the condition based on behavior duration and insulin effect.

(3) A figure depicting the temperature preference assay in Figure 1 would help illustrate the experimental approach. It is also not clear why Figure 1E is shown instead of full statistics on the individual panels shown above (the data is the same).

We have revised Figure 1A and added statistics in Figure 1BCD. We also added a figure depicting the temperature preference assay (Fig. S1).

(4) The authors state that feeding rate and amount were not changed with sucralose and glucose. However, the FLIC assay they employed does not measure consumption, so this statement is not correct, and it is unclear if the intake of sucralose and glucose is indeed comparable. This limits some of the conclusions.

We agree and removed “amount” and have revised the MS.

(5) The authors make a distinction between taste-induced and nutrient-induced warm preference. Yet the statistics in most figures only show the significance between the starved and refed flies, not the fed controls. As the recovery is in many cases incomplete and used as a distinction of nutritive vs nonnutritive signals (see Figure 1E) it will be important to also show these additional statistics to allow conclusions about how complete the recovery is.

We agree with the comments and have revised the MS and figures.

(6) The starvation period used is ranging from 1 to 3 days, as in some cases no effect was seen upon 1 day of starvation (e.g. with clock genes or temperature sensing neurons). While the authors do provide a comparison between 18-21 and 26-29 hours old flies in Figure S1, a comparison for 42-49 and 66-69 hours of starvation is missing. This also limits the conclusion as the “state” of the animal is likely quite different after 1 day vs. 3 days of starvation and, as stated by the authors, many flies die under these conditions.

We mainly used 2 overnights of starvation. Some flies (e.g. *Ilp6* mutants) were completely healthy even after 2 overnights of starvation, we had to starve them for 3 overnights. For example, *Ilp6* mutants needed 3 overnights of starvation to show a significant difference *Tp* between fed and starved flies. On the other hand, some flies (e.g. *w1118* control flies) were very sick after 2 overnights of starvation, we had to starve them for one overnight. Therefore, the starvation conditions which we used for this manuscript are from 1- 3-overnights.

First, we confirmed the starvation time by focusing on *Tp* which resulted in a statistically significant *Tp* difference between fed and starved flies; as mentioned above, flies prefer lower temperatures when starvation is prolonged (Umezaki et al., Current Biology 2018). Therefore, if *Tp* was not statistically different between fed and starved flies, we extended the starvation time from 1 to 3 overnights. Importantly, we show in Fig. S3 that the duration of starvation did not affect the recovery effect. Furthermore, since control flies do not survive 42-49 or 66-69 hours of starvation, we can not test the reviewer's suggestion. We have carefully documented the conditions in the Material and method and figure legends.

(7) In Figure 2, glucose-induced refeeding was not tested in Gr mutants or silenced animals, which would hint at post-ingestive recovery mechanisms related to nutritional intake. This is only shown later (in Figure S3) but I think it would be more fitting to address this point here. The data presented in Figure S3 regarding the taste-evoked vs nutrient-dependent warm preference is quite important while in some parts preliminary. It would nonetheless be justified to put this data in the main figures. However, some of the conclusions here are not fully supported, in part due to different and low n numbers, which due to the inherent variability of the behavior do not allow statistically sound conclusions. The authors claim that sweet GRNs are only involved in taste-induced warm preference, however, glucose is also nutritive but, in several cases, does not rescue warm preference at all upon removal of GRN function (see Figures S3A-C). This indicates that the Gal4 lines and also the involved GRs are potentially expressed in tissues/neurons required for internal nutrient sensing.

Thank you for your suggestion. We have added Figure S3ABC (glucose refeeding using Gr mutants and silenced animals) to Figure 2. There is no low N number since we tested > 5 times, i.e. >100 flies were tested. *Tp* may have a variation probably due to the effect of starvation on their temperature preference.

We did not mention that "The authors claim that sweet GRNs are only involved in taste-induced warm preference...". However, our writing may not be clear enough. We agree that "...GRs may be expressed in tissues/neurons required for internal nutrient sensing. ..." We have rewritten and revised the section.

(8) In Figure 4, fly food and glucose refeeding do not fully recover temperature preference after refeeding. With the statistical comparison to the fed control missing, this result is not consistent with the statement made in line 252. I feel this is an important point to distinguish between state-dependent and taste/nutrition-dependent changes.

We inserted the statistics and compared between Fed and other conditions.

(9) The conclusion that clock genes are required for taste-evoked warm preference is limited by the observation that they ingest less sucralose. In addition, the FLIC assay does not allow conclusions about the feeding amount, only the number of food interactions. Therefore, I think these results do not allow clear-cut conclusions about the impact of clock genes in this assay.

We agree and remove "amount" and have revised the MS. The per01 mutants ate (touched) sucralose more often than glucose. On the other hand, 1m01 mutants ate glucose more often than sucralose (Figure S6BC). However, these mutants still showed a similar TP pattern for sucralose and glucose refeeding (Fig. 5CD). The results suggest that the 1m01 flies eat enough amount of sucralose over glucose that their food intake does not affect the TP behavioral phenotype. We have rewritten and revised the section.

(10) CPR is known to be influenced by taste, thought, smell, and sight of food. As the discussion focused extensively on the CPR link to flies it would be interesting to find out whether the smell and sight of food also influence temperature preference behavior in animals with different feeding states.

We have added the data using Olfactory receptor co-receptor (Orco1) mutant, which lack olfaction, in Fig. S4. They failed to show the taste-evoked warm preference, but exhibited the nutrient-induced warm preference. Therefore, the data suggest that olfactory detection is also involved in taste-evoked warm preference. On the other hand, "seeing food" is probably more complicated, since light dramatically affects temperature preference behavior and the circadian clock that regulates temperature preference rhythms. Therefore, it will not be unlikely to draw a solid conclusion from the short set of experiments. We will address this issue in the next study.

(11) In the discussion in line 410ff the authors claim that "internal state is more likely to be associated with taste-evoked warm preference than nutrient-induced warm preference." This statement is not clear to me, as neuropeptides are involved in mediating internal state signals, both in the brain itself as well as from gut to brain. Thus, neuropeptidergic signals are also involved in nutrient-dependent state changes, the authors might just not have identified the peptides involved here. The global and developmental removal of these signals also limits the conclusions that can be drawn from the experiments, as many of these signals affect different states, circuits, and developmental progression.

We agree with the comments. We have removed the sentences and revised the MS.

Reviewer #2 (Public Review):

*Animals constantly adjust their behavior and physiology based on internal states. Hungry animals, desperate for food, exhibit physiological changes immediately upon sensing, smelling, or chewing food, known as the cephalic phase response (CPR), involving processes like increased saliva and gastrointestinal secretions. While starvation lowers body temperature, the mechanisms underlying how the sensation of food without nutrients induces behavioral responses remain unclear. Hunger stress induces changes in both behavior and physiological responses, which in flies (or at least in *Drosophila melanogaster*) leads to a preference for lower temperatures, analogous to the hunger-driven lower body temperature observed in mammals. In this manuscript, the authors have used *Drosophila melanogaster* to investigate the issue of whether taste cues can robustly trigger behavioral recovery of temperature preference in starving animals. The authors find that food detection triggers a warm preference in flies. Starved flies recover their temperature preference after food intake, with a distinction between partial and full recovery based on the duration of refeeding. Sucralose, an artificial sweetener, induces a warm preference, suggesting the importance of food-sensing cues. The paper compares the effects of sucralose and glucose refeeding, indicating that both taste cues and nutrients contribute to temperature preference recovery. The authors show that sweet gustatory receptors (Grs) and sweet GRNs (Gustatory Receptor Neurons) play a crucial role in taste-evoked warm preference. Optogenetic experiments with CsChrimson support the idea that the excitation of sweet GRNs leads to a warm preference. The authors then examine the internal state's influence on taste-evoked warm preference, focusing on neuropeptide F (NPF) and small neuropeptide F (sNPF), analogous to mammalian neuropeptide Y. Mutations in NPF and sNPF result in a failure to exhibit taste-evoked warm preference, emphasizing their role in this process. However, these neuropeptides appear not to be critical for nutrient-induced warm preference, as indicated by increased temperature preference during glucose and fly food refeeding in mutant flies. The authors also explore the role of hunger-related factors in regula3ng taste-evoked warm preference. Hunger signals, including diuretic hormone (DH44) and adipokinetic hormone (AKH) neurons, are found to be essential for taste-evoked warm preference but not for nutrient-induced warm preference. Additionally, insulin-like peptides 6 (Ilp6) and Unpaired3 (Upd3), related to nutritional stress, are identified as crucial for taste-evoked warm preference. The investigation then extends into circadian rhythms, revealing that taste-evoked warm preference does not align with the feeding rhythm. While flies exhibit a rhythmic feeding pattern, taste-evoked warm preference occurs consistently, suggesting a lack of parallel coordination. Clock genes, crucial for circadian rhythms, are found to be necessary for taste-evoked warm preference but not for nutrient-induced warm preference.*

Strengths:

A well-written and interesting study, investigating an intriguing issue. The claims, none of which to the best of my knowledge controversial, are backed by a substantial number of experiments.

Weakness:

The experimental setup used and the procedures for assessing the temperature preferences of flies are rather sparingly described. Additional details and data presentation would enhance the clarity and replicability of the study. I kindly request the authors to consider the following points:

- i) A schematic drawing or diagram illustrating the experimental setup for the temperature preference assay would greatly aid readers in understanding the spatial*

arrangement of the apparatus, temperature points, and the positioning of flies during the assay. The drawing should also be accompanied by specific details about the setup (dimensions, material, etc).

Thank you for your suggestions. We have added the schematic drawing in Fig. S1.

ii) It would be beneficial to include a visual representation of the distribution of flies within the temperature gradient on the apparatus. A graphical representation, such as a heatmaps or histograms, showing the percentage of flies within each one-degree temperature bin, would offer insights into the preferences and behaviors of the flies during the assay. In addition to the detailed description of the assay and data analysis, the inclusion of actual data plots, especially for key findings or representative trials, would provide readers with a more direct visualization of the experimental outcomes. These additions will not only enhance the clarity of the presented information but also provide the reader with a more comprehensive understanding of the experimental setup and results. I appreciate the authors' attention to these points and look forward to the potential inclusion of these elements in the revised manuscript.

Thank you for the advice. We have added the heat map for WT and Gr64fGal4>CsChrimson data in Fig. S2.

Reviewer #3 (Public Review):

Summary:

The manuscript by Yujiro Umezaki and colleagues aims to describe how taste stimuli influence temperature preference in Drosophila. Under starvation flies display a strong preference for cooler temperatures than under fed conditions that can be reversed by refeeding, demonstrating the strong impact of metabolism on temperature preference. In their present study, Umezaki and colleagues observed that such changes in temperature preference are not solely triggered by the metabolic state of the animal but that gustatory circuits and peptidergic signalling play a pivotal role in gustation-evoked alteration in temperature preference.

The study of Umezaki is definitively interesting and the findings in this manuscript will be of interest to a broad readership.

Strengths:

The authors demonstrate interesting new data on how taste input can influence temperature preference during starvation. They propose how gustatory pathways may work together with thermosensitive neurons, peptidergic neurons and finally try to bridge the gap between these neurons and clock genes. The study is very interesting and the data for each experiment alone are very convincing.

Weaknesses:

In my opinion, the authors have opened many new questions but did not fully answer the initial question - how do taste-sensing neurons influence temperature preferences? What are the mechanisms underlying this observation? Instead of jumping from gustatory neurons to thermosensitive neurons to peptidergic neurons to clock genes, the authors should have stayed within the one question they were asking at the beginning. How does sugar sensing influence the physiology of thermos-sensation in order to change temperature preference? Before addressing all the following question of the manuscript the authors should first directly decipher the neuronal interplay between these two types of neurons.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Figure S3D is cited before S2, so please rearrange the numbering.

Thank you. We have changed the numbering.

I would also suggest a different color to visualize the data points in Figure S3, as some are barely visible on the dark bars (e.g. on a dark green background).

We have revised the figures. The data points were changed to smaller opened circles.

Reviewer #2 (Recommendations For The Authors):

**Please, expand on the experimental procedure, and describe the assay in detail.*

We have added a scheme for the assay in Fig. S1 and also have revised the manuscript and figures.

**Show the distribution of the gradient data that the preference values are based upon. Not necessarily for all, but for select key experiments. Heatmaps for each replicate (stacked on top of each other) would be a nice way of showing this. Simple histograms would of course work as well.*

We have added heatmaps of selected key experiments that were added in Fig. S2. We have revised the manuscript and figures, correspondingly.

Reviewer #3 (Recommendations For The Authors)

The manuscript by Yujiro Umezaki and colleagues aims at describing how taste stimuli influence temperature preference in Drosophila. Under starvation, flies display a strong preference for cooler temperatures than under-fed conditions that can be reversed by refeeding, demonstrating the strong impact of metabolism on temperature preference. In their present study, Umezaki and colleagues observed that such changes in temperature preference are not solely triggered by the metabolic state of the animal but that gustatory circuits play a pivotal role in temperature preference. The study of Umezaki is definitively interesting and the findings in this manuscript will be of interest to a broad readership. However, I would like to draw the authors' attention to some points of concern:

The title to me sounds somehow inadequate. The definition of homeostasis (Cambridge Dictionary) is as follows: "the ability or tendency of a living organism, cell, or group to keep the conditions INSIDE it the same despite any changes in the conditions around it, or this state of internal balance". What do the authors mean by homeostatic temperature control? Reading the title not knowing much about poikilotherm insects I would understand that the authors claim that Drosophila can indeed keep a temperature homeostasis as mammals do. As Drosophila is not a homoiotherm animal and thus cannot keep its body temperature stable the title should be amended.

Homeostasis means a state of balance between all the body systems necessary for the body to survive and function properly. Drosophila are ectotherms, so the source of temperature comes from the environment, and their body temperature is very similar to that of their environment. However, the flies' temperature regulation is not simply a passive response to temperature. Instead, they actively seek a temperature based on their internal state. We have

shown that the preferred temperature increases during the day and decreases during the night, showing a circadian rhythm of temperature preference (TPR). Because their environmental temperature is very close to their body temperature, TPR gives rise to body temperature rhythms (BTR). We have shown that TPR is similar to BTR in mammals. (Kaneko et al., Current Biology 2012 and Goda et al., JBR 2023). Similarly, we showed that the hungry flies choose a lower temperature so that the body temperature is also lower. Therefore, our data suggest that the fly maintains its homeostasis by using the environmental temperature to adjust its body temperature to an appropriate temperature depending on its internal state. Therefore, I would like to keep the title as "Taste triggers a homeostatic temperature control in hungry flies" We have added more explanation in the Introduction and Discussion.

Accordingly, the authors compare the preference of flies to cooler temperatures to the reduced body temperature of mammals (Lines 64 - 65). However, according to the cited literature the reduced body temperature in starved rats is discussed to reduce metabolic heat production (Sakurada et al., 2000). The authors should more rigorously give a short summary of the findings in the cited papers and the original interpretation to help the reader not get confused.

In flies, it has been shown that a lower temperature means a lower metabolic rate, and a higher temperature means a higher metabolic rate. Therefore, hungry flies choose a lower temperature where their metabolic rate is lower and they do not need as much heat.

Similarly, in mammals, starvation causes a lower body temperature, hypothermia. Body temperature is controlled by the balance between heat loss and heat production. The starved mammals showed lower heat production. We have added this information to the introduction.

*The authors show that 5 min fly food refeeding causes a *par3al* recovery of the naïve temperature preference of the flies (Figure 1B) and that feeding of sucralose *par3ally* rescues the preference whereas glucose rescues the preference similar to refeeding with fly food would do. As glucose is both sweet and metabolically valuable it would be clearer for the reader if the authors start with the fly food experiment and then show the glucose experiment to show that the altered temperature preference depends on the food component glucose. From there they can further argue that glucose is both sweet (hedonic value) and metabolically valuable. And to disentangle sweetness from metabolism one needs a sugar that is sweet but cannot be metabolized - sucralose.*

Thank you for your advice. Since the data with sucralose is the one we want to highlight the most, we decided to present it in the order of sucralose, glucose, and fly food.

In the sucralose experiment the authors omit the 5 min data point and only show the 10 min time point. As Figure 1F indicates that both Glucose and Sucralose elicit the same attractiveness in the flies and that sweetness influences the temperature preference, it is important that the authors show the 5 min temperature preference too to underline the effect of the sweet taste stimulus on the fly behavior independent from the caloric value. Further, the authors should demonstrate not only the cumulative touches but how much sucralose or glucose may already be consumed by the fly in the depicted time frames.

It is interesting to see how much sucralose or glucose the flies consume over the time frames shown. Although the cumulative exposure to sugar is ideally equivalent to the amount of sugar, we need a different way to actually measure the amount of sugar. We will now emphasize "cumulative touches" rather than "amount of sugar" in the text. In the next study, we will look at how much sucralose or glucose the fly has already consumed.

Sucralose and Glucose have a similar molecular structure - it would be interesting to see how the sweet taste of a sugar with a different molecular structure like fructose and its receptor Gr43b (Myamoto & Amrein 2014) may contribute to temperature preferences.

Sucralose and Glucose are not structurally similar. That said, we tested fructose refeeding anyway. The hungry flies showed a taste-evoked warm preference after fructose refeeding. We have added data in Figure 1E and F. The data suggest that sweet taste is more important than sugar structure. We also tested Gr43b>CsChrimson. However, the flies do not show the taste-evoked warm preference (data not shown). The data suggest that Gr43b is not the major receptor controlling taste-evoked warm preference. We have revised the manuscript.

Both sugars appear similarly attractive to the flies (Figure 1F) - are water, sucralose, and glucose presented in a choice assay or are these individually in separate experiments?

Water, sucralose, and glucose were individually presented in separate experiments. We clarified it in the figure legend.

Subsequently, the authors address the question of how sweet taste may influence temperature preferences in flies. To this end, the authors first employ gustatory receptor mutants for Gr5a, Gr64a, and Gr61a and demonstrate that sucralose feeding does not rescue temperature preference in the absence of sweet taste receptors. In an alternative approach, the authors do not use mutants but an expression of UAS:Kir in Gr64F neurons. Taking a closer look at the graph it appears that the Kir expressing flies have an increased (nearly 1{degree sign}C) temperature preference than the starved mutant flies. Is this preference change related to the mutation directly and what would be the result if Kir would be conditionally only expressed after development is completed, or is the observed temperature preference related to the Gr64f-Gal4 line? If the latter would be the case perhaps the authors may want to bring the flies to the same genetic background to allow for a more direct comparison of the temperature preferences.

The Gr64fGal4>Kir flies show a ~one degree higher preferred temperature under starvation compared to the mutants. However, the phenotype is similar to the controls, Gr64fGal4/+ flies, under starvation. Therefore, this phenotype is not due to either the mutation or the Kir effect. Most importantly, the Gr64fGal4>Kir flies failed to show a taste-evoked warm preference. Together with other mutant data, we concluded that sweet GRNs are required for taste-evoked warm preference.

Overall, the figure legend for Figure 2 is very cryptic and should be more detailed.

We have revised the figure legend for Figure 2.

To shed light on the mechanisms underlying the changes in temperature preferences through gustatory stimuli the authors next blocked heat and cold sensing neurons in fed and starved flies and found out that TrpA1 expressing anterior cells and R11F02-Gal4 expressing neurons both participate in sweetness-induced alteration of temperature preference in starved animals. At this point, it should be explicitly indicated in the figure that the flies need more than one overnight starva3on to display the behavior (Figure 3A).

We have revised the manuscript.

The data provided by the authors indicate a kind of push-and-pull mechanism between heat and cold-sensing neurons under starvation that is somehow influenced by sweet

taste sensing. Further, the authors demonstrate that TrpA1-as well as R11F02-Gal4 driven Chrimson activation is sufficient to partially rescue temperature preference under starvation. At this point is unclear why the authors use a tubGal80ts expression system but not for the TrpA1SH-Gal4 driven Chrimson. As the development itself and the conditions under which the animals were raised may have influence on the temperature preference it is important that both groups are equally raised if the authors want to directly compare with each other.

As we wrote in the Material and Method, the R11F02-Gal4>uas-CsChrimson flies died during the development. Therefore, we had to use tubGal80ts. On the other hand, the TrpA1-Gal4>CsChrimson flies can survive to adults. As we mentioned in MS, all flies were treated with ATR after they had fully developed into adults. This means that both TrpA1-Gal4 and R11F02-Gal4 expressing cells are activated by red light via CsChrimson only in adult stages. We carefully revised the MS.

It is a pity that the authors at this point have decided to not deepen the understanding of the circuitry between thermo-sensation and metabolic homeostasis but subsequently change the focus of their study to investigate how internal state influences taste-evoked warm preference in hungry flies. Using mutants for NPF and sNPF the authors demonstrate that both peptides play a pivotal role in taste-evoked warm preference after sucrose feeding but not for nutrient-induced warm preference. Similarly, they found that DH44, AKH and dILP6, Upd2 and Upd3 neurons are also required for taste-evoked warm preference but not for nutrient-induced warm preference. Here again, the authors do not keep the systems stable and change between inhibition of neurons through Kir and mutants for peptides. For a better comparison, it would be preferable to use always exactly the same technique to inhibit neuron signalling.

It would be interesting to find the neural circuitry of thermo-sensation and metabolic homeostasis, but we do not have any luck so far. We will continue to look into the neural circuits which control taste-evoked warm preference and nutrient-induced warm preference. Since UAS-Kir is such a strong reporter, it may kill the flies sometime. So we couldn't use UAS-Kir for all Gal4 flies.

DH44 is expressed in the brain and in the abdominal ganglion where they share the expression pattern with 4 Lk neurons per hemisphere. Seeing the impact of Lk signalling in metabolism (AlAnzi et al., 2010) the authors should provide evidence that the observed effect is indeed because of DH44 and not Lk.

It would be interesting to see if Lk may play a role in taste-evoked warm preference and/or nutrient-induced warm preference. We would like to systematically screen which neuropeptides and receptors are involved in the behavior in the next study.

Seeing the results on dILP6 it is interesting that Li and Gong (2015) could show in larvae that cold-sensing neurons directly interact with dILP neurons in the brain. It would be interesting to see whether similar circuitry may exist in adult flies to regulate temperature preferences and these peptidergic neurons. Further, it appears interesting that again these animals need much longer time to display the observed shift in temperature (which again should be clearly indicated in the figure legend too). These observations should be more carefully considered in the discussion part too.

We have revised the manuscript.

In the last part of the study, the authors investigate how sensory input from temperature-sensitive cells may transmit information to central clock neurons and how

*these in turn may influence temperature preference under starvation. The experiments assume that DH44-expressing neurons play a role in the output pathway of the central clock. Using the clock gene null mutants *per* and *tim* the authors show that even though the animals display a significant starvation response neither *per* nor *tim* mutants exhibited taste-evoked warm preference, indicating a taste but not nutrient-evoked temperature preference regulation.*

The authors demonstrate interesting new data on how taste input can influence temperature preference during starvation. They propose how gustatory pathways may work together with thermosensitive neurons, peptidergic neurons and finally try to bridge the gap between these neurons and clock genes. The study is very interesting and the data for each experiment alone are very convincing. However, in my opinion, the authors have opened many new questions but did not fully answer the initial question - how do taste-sensing neurons influence temperature preferences? What are the mechanisms underlying this observation? Instead of jumping from gustatory neurons to thermosensitive neurons to peptidergic neurons to clock genes, the authors should have stayed within the one question they were asking at the beginning. How does sugar sensing influence the physiology of thermos-sensation? Before addressing all the following questions of the manuscript the authors should first directly decipher the neuronal interplay between these two types of neurons.

Thank you for your suggestion. It would be interesting to find the neural circuitry of thermosensation and metabolic homeostasis. We have tried but there is no luck so far.

The authors could e.g., employ Ca or cAMP-imaging in anterior or cold-sensitive cells and see how the responsiveness of these cells may be altered after sugar feeding. Or at least follow the idea of Li and Gong about the thermos-regulation of dILP-expressing neurons.

Thank you for your suggestion. Since we do not know how dILP-expression neurons are involved in temperature response in the adult flies. We will focus on the cells using Calcium imaging for the next study.

Anatomical analysis using the GRASP technique may further help to understand the interplay of these neurons and give new insights into the circuitry underlying food preference alteration under starvation.

Thank you for your suggestion. It would be interesting to find the neural circuitry of thermosensation and metabolic homeostasis. We have tried but there is no luck so far.

Minor comments:

Line 51: Hungry animals are desperate for food - I think the authors should not anthropomorphize at this point too much but rather strictly describe how the animals change their behavior without any interpretation of the mental state of the animal.

We have modified the manuscript.

Line 80: Hunger and satiety dramatically affect animal behavior and physiology and control feeding - please not only cite the papers but also give a short overview of the cited papers on which behaviors are altered and how.

We have revised the manuscript.

Overall statistic: The authors do comparative statistics always against starved animals throughout but often state in the text a comparison against fed (Line 111: "but did not

| *reach that of the fed flies") I think the authors should describe the date according to their statistics and keep this constant throughout the paper.*

Sorry for the confusion. We originally had it, but we removed it. We have added the additional statistical analyses.

| *Figure legends: Overall the figure legends could be more developed and more detailed.*

We have revised the manuscript.

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